

Chapter 6 – Maneuvering Models for Ships and USVs

An alternative to the seakeeping formalism (Chapter 5) is to use maneuvering theory to describe the motions of ships and USVs. In classical maneuvering theory, the frequency-dependent added-mass and radiation-damping matrices are replaced by constant hydrodynamic coefficients, typically chosen as their zero-frequency values for horizontal-plane motions in surge, sway, and yaw. Hence, the fluid-memory effects are neglected, and the equations can be formulated directly in the body-fixed reference frame.

For fully coupled 6-DOF maneuvering models, constant equivalent added-mass and damping matrices can alternatively be obtained by power-based spectral averaging of the frequency-dependent seakeeping coefficients. This yields a physically consistent first-principles nonlinear mass–damper–spring system suitable for simulation, guidance, navigation, and control applications.



Chapter Goals

- Formulate the **first-principles 6-DOF maneuvering model** by combining rigid-body kinetics with hydrodynamic added mass, damping, hydrostatic restoring, and external forces.
- Explain how the rigid-body and added-mass **Coriolis–centripetal matrices** arise in **BODY-fixed coordinates** and state their skew-symmetry property.
- Explain when **zero-frequency potential coefficients** are appropriate for surge, sway, and yaw, why **viscous damping must be added**, and why the approximation fails for coupled 6-DOF motion.
- Apply the **power-based spectral averaging method** to compute sea-state-dependent constant equivalent added-mass and damping matrices for **6-DOF maneuvering models**.
- Model linear and nonlinear hydrodynamic damping using **ITTC surge resistance, cross-flow drag**, and additional **roll, pitch, and yaw damping**.
- Include **constant irrotational ocean currents** using relative velocity and formulate the dynamics with either relative or absolute velocity as the state.
- Derive and compare **simplified and classical maneuvering models**, including longitudinal–lateral, horizontal-plane 3-DOF, dynamic positioning, Abkowitz, second-order modulus, and MMG models.

$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + C_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

First-Principles 6-DOF Maneuvering Model

Chapter 6 Summary: First-Principles 6-DOF Maneuvering Model

Maneuvering Model including Ocean Currents

$$\underbrace{M_{RB}\dot{\mathbf{v}} + C_{RB}(\mathbf{v})\mathbf{v}}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\mathbf{v}}_r + C_A(\mathbf{v}_r)\mathbf{v}_r + D(\mathbf{v}_r)\mathbf{v}_r + d(\mathbf{v}_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$M = M_{RB} + M_A$ – system inertia matrix (including added mass)

$C(\mathbf{v}_r) = C_{RB}(\mathbf{v}_r) + C_A(\mathbf{v}_r)$ – Coriolis-centripetal matrix (including added mass)

$D(\mathbf{v}_r)$ – hydrodynamic damping matrix

$d(\mathbf{v}_r)$ – nonlinear dissipative forces and moments not expressible in the form $D(\mathbf{v}_r)\mathbf{v}_r$

$g(\boldsymbol{\eta})$ – vector of gravitational and buoyancy forces and moments

$\boldsymbol{\tau}$ – vector of control forces and moments

$\boldsymbol{\tau}_{\text{wind}}$ – vector of wind-induced forces and moments

$\boldsymbol{\tau}_{\text{wave}}$ – vector of wave-induced forces and moments

Property 10.1 (Irrotational constant ocean currents)

The maneuvering model can be expressed entirely in terms of the relative velocity vector $\mathbf{v}_r = \mathbf{v} - \mathbf{v}_c$. If the rigid-body Coriolis and centripetal matrix $C_{RB}(\mathbf{v})$ is parametrized independent of linear velocity $\mathbf{v}_1 = [u, v, w]^T$ (Section 3.3.2) and the ocean current

$$\mathbf{v}_c = \underbrace{[u_c, v_c, w_c, 0, 0, 0]^T}_{\mathbf{v}_c^b}$$

is irrotational and constant. Hence,

$$M_{RB}\dot{\mathbf{v}} + C_{RB}(\mathbf{v})\mathbf{v} = M_{RB}\dot{\mathbf{v}}_r + C_{RB}(\mathbf{v}_r)\mathbf{v}_r$$

This implies that

$$M = M_{RB} + M_A$$

$$C(\mathbf{v}_r) = C_{RB}(\mathbf{v}_r) + C_A(\mathbf{v}_r)$$

$$M\dot{\mathbf{v}}_r + C(\mathbf{v}_r)\mathbf{v}_r + D(\mathbf{v}_r)\mathbf{v}_r + d(\mathbf{v}_r) + g(\boldsymbol{\eta}) = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

Chapter 6 Summary: First-Principles 6-DOF Maneuvering Model

Two state-space formulations are commonly used. The first uses the relative velocity $\boldsymbol{\nu}_r$ as state and is convenient when ocean currents are modeled explicitly and relative velocity is measured (e.g., by a DVL). The second uses the absolute velocity $\boldsymbol{\nu}$ as state and is often preferred for surface vessels, where INS/GNSS systems provide measurements of absolute velocity.

Relative-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}) (\boldsymbol{\nu}_r + \boldsymbol{\nu}_c)$$

$$\dot{\boldsymbol{\nu}}_r = \mathbf{M}^{-1} (\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r) - \mathbf{g}(\boldsymbol{\eta}))$$

$$\boldsymbol{\nu}_c = \underbrace{[u_c, v_c, w_c, 0, 0, 0]^T}_{\mathbf{v}_c^b}$$

$$\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$$

Absolute-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}) \boldsymbol{\nu}$$

$$\dot{\boldsymbol{\nu}} = \dot{\boldsymbol{\nu}}_c + \mathbf{M}^{-1} (\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r) - \mathbf{g}(\boldsymbol{\eta}))$$

$$\mathbf{v}_c^b = [u_c, v_c, w_c]^T$$

$$\mathbf{v}_c^n = \mathbf{R}(\boldsymbol{\Theta}_{nb}) \mathbf{v}_c^b$$

$$\dot{\boldsymbol{\nu}}_c = \begin{bmatrix} -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Definition 10.1 (Irrotational constant ocean current)

An irrotational ocean current that is constant in the inertial frame $\{n\}$ satisfies

$$\dot{\mathbf{v}}_c^n = \dot{\mathbf{R}}(\boldsymbol{\Theta}_{nb}) \mathbf{v}_c^b + \mathbf{R}(\boldsymbol{\Theta}_{nb}) \dot{\mathbf{v}}_c^b := \mathbf{0}$$

where $\dot{\mathbf{R}}(\boldsymbol{\Theta}_{nb}) = \mathbf{R}(\boldsymbol{\Theta}_{nb}) \mathbf{S}(\boldsymbol{\omega}_{nb}^b)$. Consequently,

$$\dot{\mathbf{v}}_c^b = -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b.$$

Rigid-Body Kinetics

Newton-Euler Equations (Chapter 3)

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{hyd}} + \boldsymbol{\tau}_{\text{hs}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bG}^b) \\ m\mathbf{S}(\mathbf{r}_{bG}^b) & \mathbf{I}_b^b \end{bmatrix}$$

$$= \begin{bmatrix} m & 0 & 0 & 0 & mz_G & -my_G \\ 0 & m & 0 & -mz_G & 0 & mx_G \\ 0 & 0 & m & my_G & -mx_G & 0 \\ 0 & -mz_G & my_G & I_x & -I_{xy} & -I_{xz} \\ mz_G & 0 & -mx_G & -I_{yx} & I_y & -I_{yz} \\ -my_G & mx_G & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}$$

Linear velocity-independent parametrization.

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_b^b \boldsymbol{\nu}_2) \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b)$$

$$= \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b)$$

$$\mathbf{v} = [u, v, w, p, q, r]^\top$$

$$\boldsymbol{\nu}_1 := \mathbf{v}_{nb}^b = [u, v, w]^\top$$

$$\boldsymbol{\nu}_2 := \boldsymbol{\omega}_{nb}^b = [p, q, r]^\top$$

$$\mathbf{M}_{RB} = \mathbf{M}_{RB}^\top > 0$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^\top(\boldsymbol{\nu})$$

Screw transformation between two points

$$\boldsymbol{\nu}_{nP} = \mathbf{H}(\mathbf{r}_{bP}^b) \boldsymbol{\nu}$$

$$\mathbf{H}(\mathbf{r}_{bP}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bP}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}$$

$$\mathbf{H}^\top(\mathbf{r}_{bP}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bP}^b) & \mathbf{I}_3 \end{bmatrix}$$

Hydrodynamic Derivatives for Added Mass and Linear Damping

The hydrodynamic forces and moments acting on the body are modeled as

$$\boldsymbol{\tau}_{\text{hyd}} = -\mathbf{M}_A \dot{\boldsymbol{\nu}}_r - \mathbf{C}_A(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r - \mathbf{d}(\boldsymbol{\nu}_r)$$

$$\mathbf{M}_A = \mathbf{M}_A^\top \geq 0, \\ \mathbf{D} > 0$$

Added Mass Matrix

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

Notation of SNAME (1950)
for Hydrodynamic Derivatives

$$Y = -Y_{\dot{u}} \dot{u}, \quad Y_{\dot{u}} := \frac{\partial Y}{\partial \dot{u}}$$

$$M_{A,21} = -Y_{\dot{u}}$$

Linear Damping Matrix

$$\mathbf{D} = - \begin{bmatrix} X_u & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r \end{bmatrix}$$

$$D_{66} = -N_r$$

$$N = -N_r r, \quad N_r := \frac{\partial N}{\partial r}$$

Estimation from Zero-Frequency Seakeeping Coefficients

3-DOF Zero-Frequency Approximation (Classical Maneuvering Theory)

Zero-Frequency added mass and damping matrices can be used for surge, sway, and yaw motions (DOF 1–2–6), since radiation vanishes at high frequency. Note that when a vessel oscillates in waves, it radiates outgoing waves.

Added mass matrix: $M_{A,ij} \approx A_{ij}(0)$

Linear damping matrix: $D_{ij} \approx B_{ij}(0) + B_{v,ij}$

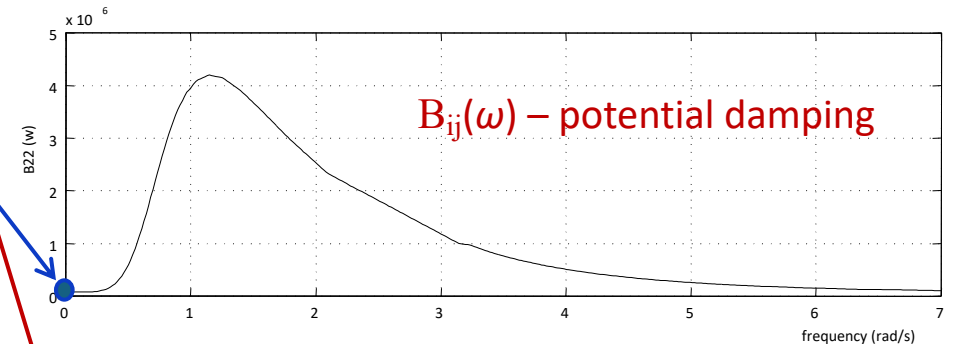
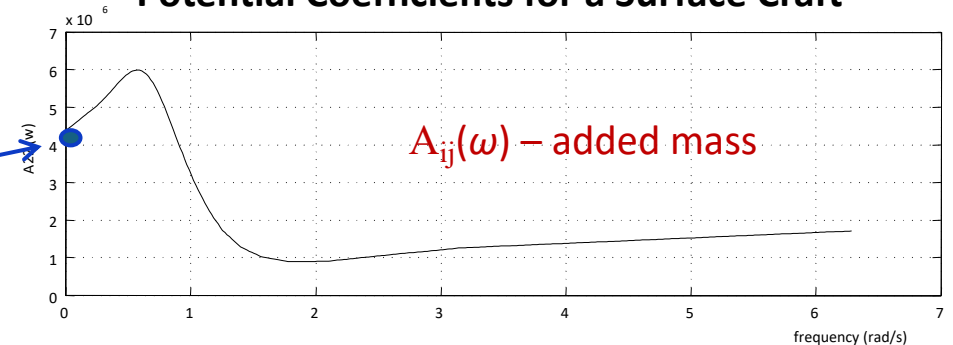
Should not be extended to heave, roll, and pitch (DOF 3–4–5):

- These motions generate strong radiation damping since vertical oscillations radiate waves and dissipate energy.
- Their natural frequencies lie in the wave-frequency range. Hence, the coefficients are strongly frequency dependent.
- Using only zero-frequency values leads to incorrect energy dissipation and unrealistic motions.

Problem that coupling terms (A_{13} , A_{15} , A_{24} , A_{26} , A_{35} , A_{46}) peak at different frequencies:

- The zero frequency cannot represent these mixed-frequency effects.
- **Solution:** spectral averaging, such as the power-based averaging method (see next page).

Potential Coefficients for a Surface Craft

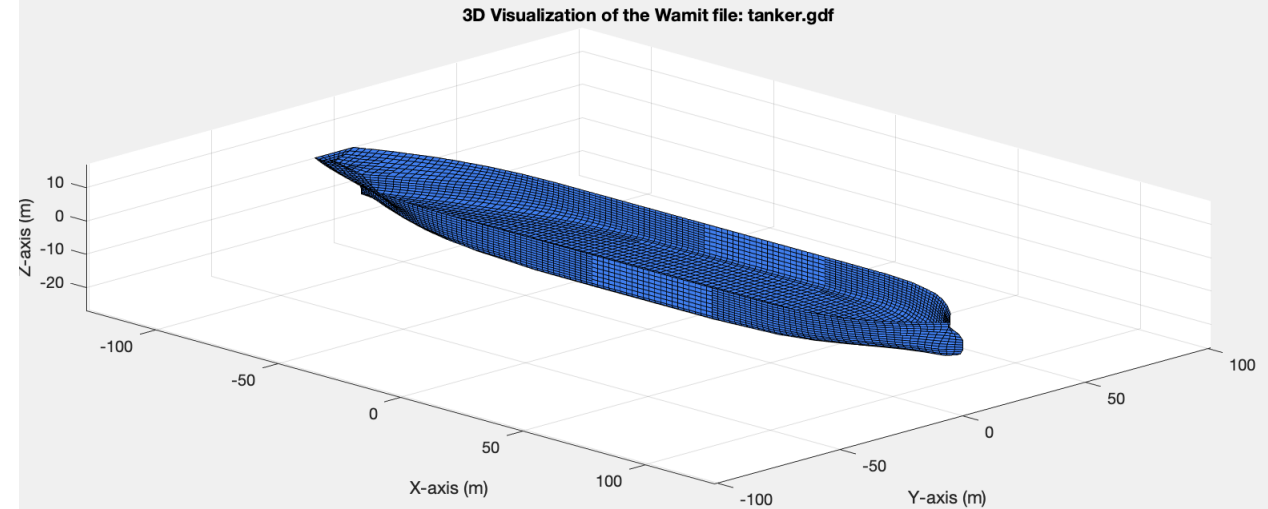


Viscous damping B_v must be added since potential damping $B(0)$ is zero. Alternatively, D can be estimated from time constants/relative damping ratios.

MSS Toolbox

```
>> load tanker
>> vessel
vessel =
  struct with fields:
    main: [1x1 struct]
    velocities: 0
    headings: [1x36 double]
    MRB: [6x6 double]
    C: [6x6x60 double]
    freqs: [1x60 double]
    A: [6x6x60 double]
    B: [6x6x60 double]
    motionRAO: [1x1 struct]
    forceRAO: [1x1 struct]
    driftfr: [1x1 struct]
    Bv: [6x6 double]
>> vessel.freqs
ans =
Columns 1 through 8
  0    0.0383    0.0424    0.0468    0.0517    0.0571    0.0631    0.0697
Columns 9 through 16
  0.0770    0.0850    0.0939    0.1037    0.1145    0.1264    0.1395    0.1540
Columns 17 through 24
  0.1700    0.1876    0.2070    0.2284    0.2519    0.2779    0.3063    0.3378
Columns 25 through 32
  0.3722    0.4101    0.4517    0.4971    0.5473    0.6018    0.6621    0.7281
Columns 33 through 40
  0.7994    0.8775    0.9622    1.0560    1.1571    1.2668    1.3840    1.5104
Columns 41 through 48
  1.6491    1.8003    1.9574    2.1299    2.3100    2.5033    2.7083    2.9361
Columns 49 through 56
  3.1574    3.3963    3.6530    3.9270    4.1888    4.4880    4.7600    5.0671
Columns 57 through 60
  5.3702    5.7120    5.9840    10.0000
```

```
>> plot_wamitgdf('tanker')
```



```
>> MA = vessel.A(:,:,1)
MA =
1.0e+11 *
  0.0000    0    -0.0000    0    -0.0128    0
  0    0.0005    0    0.0030    0    0.0029
 -0.0000    0    0.0039    0    0.0096    0
  0    0.0030    0    0.1103    0    -0.0419
 -0.0128    0    0.0096    0    9.0021    0
  0    0.0029    0    -0.0420    0    1.7383
```

Zero-frequency added mass matrix

Estimation from Power-Averaged Seakeeping Coefficients

For a fully coupled 6-DOF maneuvering model, the zero-frequency approximation is not appropriate. The zero frequency cannot represent the coupling terms (A_{13} , A_{15} , A_{24} , A_{26} , A_{35} , A_{46}) which peak at different frequencies. However, spectral averaging solves this problem.

6-DOF Equivalent Matrices by Power-Based Averaging (Section 5.4)

Frequency-dependent added mass $\mathbf{A}_U(\omega)$ and potential damping $\mathbf{B}_U(\omega)$ for constant speed U are integrated over the full frequency band relevant to a given sea state, weighted by the spectral energy distribution using a normalized wave spectrum to ensure that only the spectral shape influences the equivalent values $\mathbf{A}_U^{\text{eq}}$ and $\mathbf{B}_U^{\text{eq}}$ (Fossen, 2025):

$$\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}, \quad \mathbf{D} = \mathbf{B}_0^{\text{eq}} + \mathbf{B}_v$$

In practical applications, the zeroth spectral moment m_0 is evaluated for $U = 0$, thereby avoiding explicit dependence of the weighting spectrum on the forward speed U and relative wave heading β_{rel} such that $\omega_e = \omega$. The peak frequency ω_p is then treated as a constant parameter describing the average sea state for the operating condition under consideration.

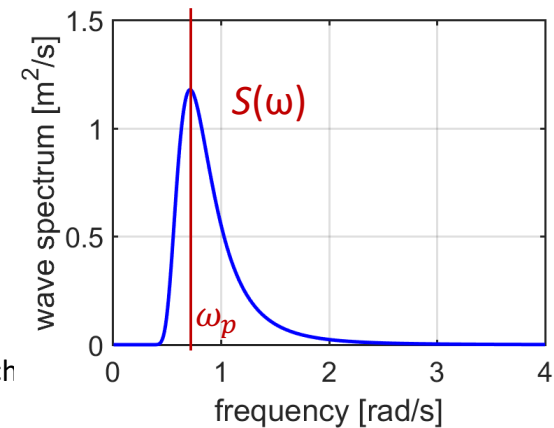
$$\mathbf{A}_U^{\text{eq}} \approx \frac{1}{m_0} \sum_{k=1}^N \mathbf{A}_U(\omega[k]) S(\omega[k]) \Delta\omega$$

$$\mathbf{B}_U^{\text{eq}} \approx \frac{1}{m_0} \sum_{k=1}^N \mathbf{B}_U(\omega[k]) S(\omega[k]) \Delta\omega.$$

PM-spectrum used for weighting

$$S(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left(-\beta \left(\frac{\omega_p}{\omega}\right)^4\right) \quad S^N(\omega) = \frac{S(\omega)}{m_0}$$

$$m_0 = \int_0^{\omega_{\max}} S(\omega) d\omega \approx \sum_{k=1}^N S(\omega[k]) \Delta\omega.$$



Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-Dependent Added Mass and Damping: A Power-Based Approach Oceanic Engineering 341, 122494. <https://doi.org/10.1016/j.oceaneng.2025.122494>

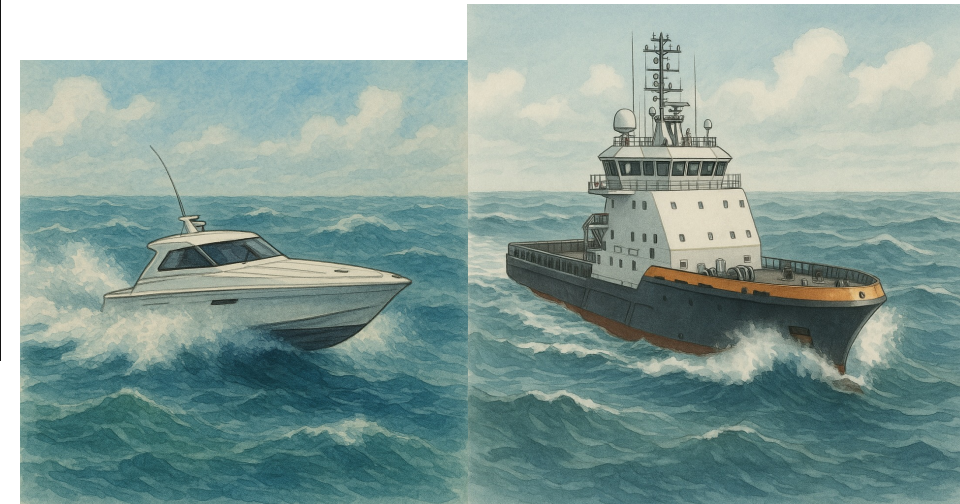
MSS Toolbox

```
load supply;
omega_p = 0.8;
vessel = computeManeuveringModel(vessel, omega_p, 0);
```

```
function vessel = computeManeuveringModel(vessel, omega_p, plotFlag)
% computeManeuveringModel is compatible with MATLAB and GNU Octave (www.octave.org)
% Computes the equivalent added mass vessel.A_eq(:, :, k, n) and damping
% vessel.B_eq(:, :, k, n) matrices for each omega_p(k) and velocity(n) by integrating
% the frequency-dependent hydrodynamic matrices A_U(omega) and B_U(omega) using the wave
% spectrum S(omega) as a weighting function.
%
% The equivalent matrices are computed as:
%
%  $A_{eq} = \int A_U(\omega) S_N(\omega) d\omega$ 
%  $B_{eq} = \int B_U(\omega) S_N(\omega) d\omega$ 
%
% where
%
% S(omega) : Wave energy spectrum
% S_N(omega) = S(omega) / m_0 : Normalized wave spectrum
% m_0 =  $\int S(\omega) d\omega$  : Zero spectral moment
% A_U(omega) and B_U(omega) : Added mass and damping matrices at speed U
%
% The integrals are evaluated numerically using trapezoidal integration.
% This ensures that the kinetic energy and power dissipation properties
% of the frequency-dependent system are preserved in the equivalent
% constant matrices.
```

```
>> vessel.A_eq
ans =
  1.0e+09 *
    0.0014         0         0         0         0         0
         0    0.0047         0    0.0063         0   -0.0084
         0         0    0.0088         0    0.1080         0
         0    0.0063         0    0.0914         0   -0.1839
         0         0    0.1080         0    4.8111         0
         0   -0.0084         0   -0.1839         0    2.0879

>> vessel.B_eq
ans =
  1.0e+09 *
    0.0009         0         0         0         0         0
         0    0.0029         0    0.0038         0   -0.0011
         0         0    0.0061         0    0.0873         0
         0    0.0038         0    0.0197         0   -0.0781
         0         0    0.0873         0    4.1030         0
         0   -0.0011         0   -0.0781         0    0.8871
```



Estimation from Time Constants and Relative Damping Ratios

The response of a marine craft can be approximated by (Section 6.1.1):

- Time constants T_i (1st-order systems).
- Relative damping ratios ζ_i (2nd-order systems).

Assume that the added mass is approximated by a constant diagonal matrix about the CG using semi-empirical formulas:

$$M_A = \text{diag} \{M_{A,11}, M_{A,22}, M_{A,33}, M_{A,44}, M_{A,55}, M_{A,66}\}$$

$$D_{11} = \frac{m + M_{A,11}}{T_1^*}, \quad D_{22} = \frac{m + M_{A,22}}{T_2^*}, \quad D_{66} = \frac{I_z + M_{A,66}}{T_6^*}.$$

Motion	Time Constant	
Surge	$T_1^* = 5\text{--}50 \text{ s}$	■ Small craft, USVs, and AUVs: lower values
Sway	$T_2^* = 1\text{--}20 \text{ s}$	■ Ships and offshore vessels: higher values
Yaw	$T_6^* = 1\text{--}30 \text{ s}$	

$$D_{33} = 2\zeta_3^* \omega_3 (m + M_{A,33})$$

$$D_{44} = 2\zeta_4^* \omega_4 (I_x + M_{A,44})$$

$$D_{55} = 2\zeta_5^* \omega_5 (I_y + M_{A,55})$$

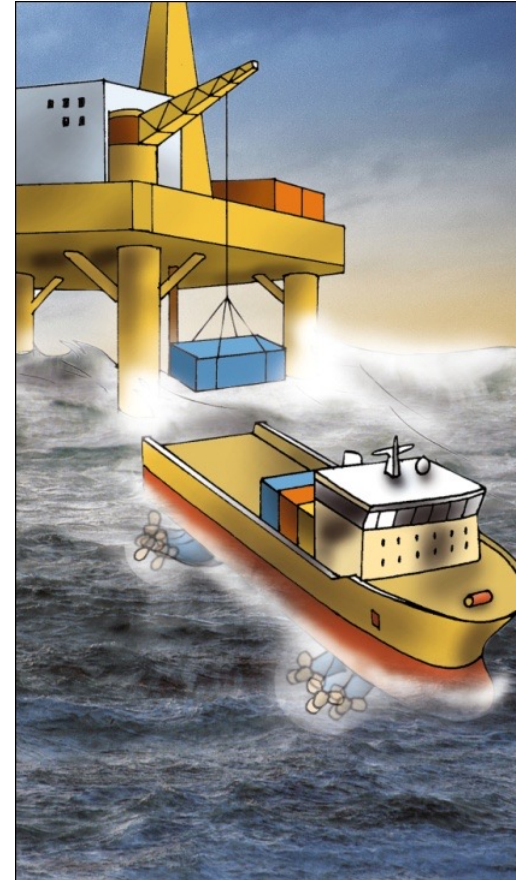
Motion	Relative Damping Ratio
Heave	$\zeta_3^* = 0.1\text{--}0.3$
Roll	$\zeta_4^* = 0.05\text{--}1.0$
Pitch	$\zeta_5^* = 0.1\text{--}0.4$

$$D \approx \begin{bmatrix} D_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & D_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & D_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & D_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & D_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & D_{66} \end{bmatrix}$$

- Streamlined AUVs and torpedo-shaped vehicles: lower values
- Ships and offshore vessels: higher values

MSS Toolbox

```
function D = Dmtrx(T_126,zeta_45,MRB,MA,hydrostatics)
% D = Dmtrx([T1, T2, T6],[zeta4,zeta5],MRB,MA,hydrostatics)
% computes the 6x6 linear damping matrix for marine craft (submerged and
% floating) by specifying the time constants [T1, T2, T6] in DOFs 1,2 and 6.
% The time constants can be found by open-loop step responses. For roll and
% pitch the relative damping ratios are specified using [zeta4, zeta5].
% For floating vessels it is assumed that zeta3 = 0.2 in heave, while
% submerged vehicles are assumed to be neutrally buoyant,  $W = B$ , with equal
% time constants in heave and sway, that is  $T3 = T2$ .
%
% Inputs: T_126 = [T1, T2, T6]: time constants for DOFs 1, 2 and 6
%         zeta_45 = [zeta4, zeta5]: relative damping ratios in DOFs 4 and 5
%         MRB: 6x6 rigid-body system matrix (see rbody.m)
%         MA: 6x6 hydrodynamic added mass system matrix
%         hydrostatics = G for surface craft (see Gmtrx.m)
%         hydrostatics = [W r_bg' r_bb'] for neutrally buoyant submerged
%         vehicles where  $W = m \cdot g$ ,  $r_{bg} = [xg, yg, zg]'$  and  $r_{bb} = [xb, yb, zb]'$ 
%
% Output: D: 6x6 diagonal linear damping matrix
%
% Examples:
% supply: D = Dmtrx([10 50 20], [0.2, 0.3], MRB, MA, G)
% tanker: D = Dmtrx([20 100 50], [0.2, 0.3], MRB, MA, G)
% AUV:    D = Dmtrx([5 10 10], [0.2, 0.3], MRB, MA, [m*g, [0,0,zg], [0,0,0]])
```



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Coriolis–Centripetal Forces in a Rotating Coordinate System

Definition 5.1 (Added mass) *Hydrodynamic added mass is the apparent increase in inertia caused by the need to accelerate the surrounding fluid together with the vessel. Hence, an accelerating vessel behaves as if its rigid-body mass and inertia are larger than their physical values.*

In Section 3.1, the rigid-body kinetics of a marine craft were derived using Newton's laws. In an analogous manner, the hydrodynamic added-mass forces and moments can be separated into terms associated with the added-mass system inertia matrix M_A and terms associated with the added-mass Coriolis–centripetal matrix $C_A(\nu)$. To derive the expression for $C_A(\nu)$, an energy-based approach using Kirchhoff's equations will now be presented.

The goal is to derive a formula for the matrix $C_A(\nu_r)$ such that it can be computed as a function of the matrix $M_A = A_0^{\text{eq}}$

$$\underbrace{M_{RB}\dot{\nu} + C_{RB}(\nu)\nu}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\nu}_r + C_A(\nu_r)\nu_r + D(\nu_r)\nu_r + d(\nu_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\eta)}_{\text{hydrostatic forces}} = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

The added mass matrix M_A should not be interpreted as a fixed amount of water being carried or dragged along with the vessel. Such an interpretation would correspond to a constant increase in mass.

Instead, added mass is a dynamic effect arising from the pressure field generated by the accelerating motion, and its value therefore depends on the geometry of the body and the oscillation frequency.

The body behaves as if its effective mass is the sum of its rigid-body mass and the added mass

$$M = M_{RB} + M_A$$

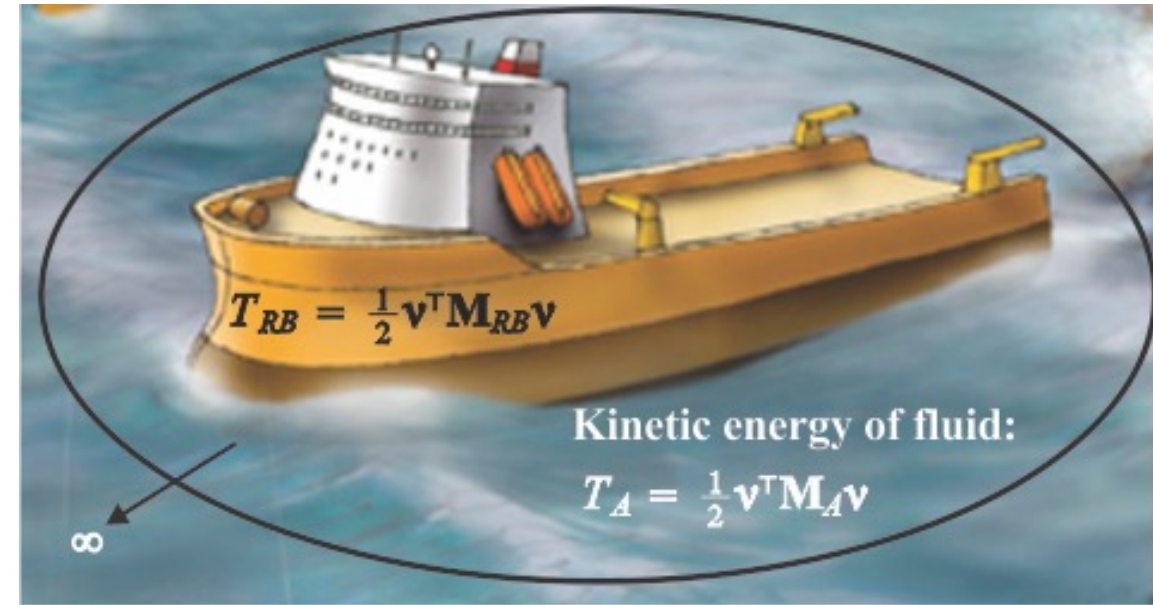
Coriolis–Centripetal Forces in a Rotating Coordinate System

Fluid Kinetic Energy

- The concept of fluid kinetic energy can be used to derive the added mass Coriolis–centripetal matrix $C_M(\Lambda_v)$
- Any motion of the vessel will induce a motion in the otherwise stationary fluid. To allow the vessel to pass through the fluid, it must move aside and then close behind the vessel.
- Consequently, the fluid motion possesses kinetic energy that it would lack otherwise (Lamb 1932).

$$T_A = \frac{1}{2} \boldsymbol{\nu}^T \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$



Kinetic energy of the surrounding water

When the hull moves with BODY velocity $\mathbf{v}$, it induces an irrotational flow in an unbounded domain. Consider a control volume that encloses the ship and extends to a far field surface S_∞ (water at rest as $r \rightarrow \infty$).

The kinetic energy of this induced flow is

$$T_A = \frac{1}{2} \boldsymbol{\nu}^T \mathbf{M}_A \boldsymbol{\nu}$$

so the total kinetic energy is (ship-water system)

$$T = \frac{1}{2} \boldsymbol{\nu}^T (\mathbf{M}_{RB} + \mathbf{M}_A) \boldsymbol{\nu}.$$

Coriolis–Centripetal Forces in a Rotating Coordinate System

A formula for the matrix $\mathbf{C}_A(\mathbf{v}_r)$ as a function of the matrix $\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}$ can be derived using Kirchoff's equations.

Euler-Lagrange's Equation (Only for Generalized Coordinates)

$L = T - V$ Difference between kinetic and potential energy

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\boldsymbol{\eta}}} \right) - \frac{\partial L}{\partial \boldsymbol{\eta}} = \mathbf{J}_{\Theta}^{-\top}(\boldsymbol{\eta}) \boldsymbol{\tau}$$

6-DOF Generalized Coordinates:

$$\boldsymbol{\eta} = [x^n, y^n, z^n, \phi, \theta, \psi]^{\top}$$

Quaternions are not generalized coordinates:

$$\boldsymbol{\eta} = [x^n, y^n, z^n, \eta, \varepsilon_1, \varepsilon_2, \varepsilon_3]^{\top}$$

Kirchhoff's Equations

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_1} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_1 \\ \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_2} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_2} + \mathbf{S}(\boldsymbol{\nu}_1) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_2 \end{aligned}$$

$$\mathbf{v}_1 = [u, v, w]^{\top} \quad \boldsymbol{\tau}_1 = [X, Y, Z]^{\top}$$

$$\mathbf{v}_2 = [p, q, r]^{\top} \quad \boldsymbol{\tau}_2 = [K, M, N]^{\top}$$

$$T_A = \frac{1}{2} \boldsymbol{\nu}^{\top} \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

Fluid kinetic energy

Coriolis–Centripetal Forces in a Rotating Coordinate System

Kirchhoff's Equations

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_1} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_1 \\ \frac{d}{dt} \left(\frac{\partial T_A}{\partial \boldsymbol{\nu}_2} \right) + \mathbf{S}(\boldsymbol{\nu}_2) \frac{\partial T_A}{\partial \boldsymbol{\nu}_2} + \mathbf{S}(\boldsymbol{\nu}_1) \frac{\partial T_A}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_2 \end{aligned}$$

$$T_A = \frac{1}{2} \boldsymbol{\nu}^\top \mathbf{M}_A \boldsymbol{\nu}, \quad \dot{\mathbf{M}}_A = \mathbf{0}$$

Property 6.1 (Added-mass system inertia matrix $\mathbf{M}_A$)

For a rigid body at rest or moving at constant forward speed $U \geq 0$ in an ideal fluid, the added-mass system inertia matrix is symmetric and positive semidefinite:

$$\mathbf{M}_A = \mathbf{M}_A^\top \geq \mathbf{0}.$$

$$\mathbf{M}_A = - \begin{bmatrix} X_{\ddot{u}} & X_{\ddot{v}} & X_{\ddot{w}} & X_{\ddot{p}} & X_{\ddot{q}} & X_{\ddot{r}} \\ Y_{\ddot{u}} & Y_{\ddot{v}} & Y_{\ddot{w}} & Y_{\ddot{p}} & Y_{\ddot{q}} & Y_{\ddot{r}} \\ Z_{\ddot{u}} & Z_{\ddot{v}} & Z_{\ddot{w}} & Z_{\ddot{p}} & Z_{\ddot{q}} & Z_{\ddot{r}} \\ K_{\ddot{u}} & K_{\ddot{v}} & K_{\ddot{w}} & K_{\ddot{p}} & K_{\ddot{q}} & K_{\ddot{r}} \\ M_{\ddot{u}} & M_{\ddot{v}} & M_{\ddot{w}} & M_{\ddot{p}} & M_{\ddot{q}} & M_{\ddot{r}} \\ N_{\ddot{u}} & N_{\ddot{v}} & N_{\ddot{w}} & N_{\ddot{p}} & N_{\ddot{q}} & N_{\ddot{r}} \end{bmatrix}$$

Component form

$$\begin{aligned} X_A &= X_{\ddot{u}}\ddot{u} + X_{\ddot{w}}(\ddot{w} + uq) + X_{\ddot{q}}\ddot{q} + Z_{\ddot{w}}wq + Z_{\ddot{q}}q^2 \\ &\quad + X_{\ddot{v}}\ddot{v} + X_{\ddot{p}}\ddot{p} + X_{\ddot{r}}\ddot{r} - Y_{\ddot{v}}vr - Y_{\ddot{p}}rp - Y_{\ddot{r}}r^2 \\ &\quad - X_{\ddot{v}}ur - Y_{\ddot{w}}wr \\ &\quad + Y_{\ddot{w}}vq + Z_{\ddot{p}}pq - (Y_{\ddot{q}} - Z_{\ddot{r}})qr \\ Y_A &= X_{\ddot{v}}\ddot{u} + Y_{\ddot{w}}\ddot{w} + Y_{\ddot{q}}\ddot{q} \\ &\quad + Y_{\ddot{v}}\ddot{v} + Y_{\ddot{p}}\ddot{p} + Y_{\ddot{r}}\ddot{r} + X_{\ddot{v}}vr - Y_{\ddot{w}}vp + X_{\ddot{r}}r^2 + (X_{\ddot{p}} - Z_{\ddot{r}})rp - Z_{\ddot{p}}p^2 \\ &\quad - X_{\ddot{w}}(up - wr) + X_{\ddot{u}}ur - Z_{\ddot{w}}wp \\ &\quad - Z_{\ddot{q}}pq + X_{\ddot{q}}qr \\ Z_A &= X_{\ddot{w}}(\ddot{u} - wq) + Z_{\ddot{w}}\ddot{w} + Z_{\ddot{q}}\ddot{q} - X_{\ddot{u}}uq - X_{\ddot{q}}q^2 \\ &\quad + Y_{\ddot{w}}\ddot{v} + Z_{\ddot{p}}\ddot{p} + Z_{\ddot{r}}\ddot{r} + Y_{\ddot{v}}vp + Y_{\ddot{r}}rp + Y_{\ddot{p}}p^2 \\ &\quad + X_{\ddot{v}}up + Y_{\ddot{w}}wp \\ &\quad - X_{\ddot{v}}vq - (X_{\ddot{p}} - Y_{\ddot{q}})pq - X_{\ddot{r}}qr \\ K_A &= X_{\ddot{p}}\ddot{u} + Z_{\ddot{p}}\ddot{w} + K_{\ddot{q}}\ddot{q} - X_{\ddot{v}}wu + X_{\ddot{r}}uq - Y_{\ddot{w}}w^2 - (Y_{\ddot{q}} - Z_{\ddot{r}})wq + M_{\ddot{r}}r^2 \\ &\quad + Y_{\ddot{p}}\ddot{v} + K_{\ddot{p}}\ddot{p} + K_{\ddot{r}}\ddot{r} + Y_{\ddot{w}}v^2 - (Y_{\ddot{q}} - Z_{\ddot{r}})vr + Z_{\ddot{p}}vp - M_{\ddot{r}}r^2 - K_{\ddot{q}}rp \\ &\quad + X_{\ddot{w}}uv - (Y_{\ddot{v}} - Z_{\ddot{w}})vw - (Y_{\ddot{r}} + Z_{\ddot{q}})wr - Y_{\ddot{p}}wp - X_{\ddot{q}}ur \\ &\quad + (Y_{\ddot{r}} + Z_{\ddot{q}})vq + K_{\ddot{r}}pq - (M_{\ddot{q}} - N_{\ddot{r}})qr \\ M_A &= X_{\ddot{q}}(\ddot{u} + wq) + Z_{\ddot{q}}(\ddot{w} - uq) + M_{\ddot{q}}\ddot{q} - X_{\ddot{w}}(u^2 - w^2) - (Z_{\ddot{w}} - X_{\ddot{u}})wu \\ &\quad + Y_{\ddot{q}}\ddot{v} + K_{\ddot{q}}\ddot{p} + M_{\ddot{r}}\ddot{r} + Y_{\ddot{p}}vr - Y_{\ddot{r}}vp - K_{\ddot{r}}(p^2 - r^2) + (K_{\ddot{p}} - N_{\ddot{r}})rp \\ &\quad - Y_{\ddot{w}}uv + X_{\ddot{v}}vw - (X_{\ddot{r}} + Z_{\ddot{p}})(up - wr) + (X_{\ddot{p}} - Z_{\ddot{r}})(wp + ur) \\ &\quad - M_{\ddot{r}}pq + K_{\ddot{q}}qr \\ N_A &= X_{\ddot{r}}\ddot{u} + Z_{\ddot{r}}\ddot{w} + M_{\ddot{r}}\ddot{q} + X_{\ddot{v}}u^2 + Y_{\ddot{w}}wu - (X_{\ddot{p}} - Y_{\ddot{q}})uq - Z_{\ddot{p}}wq - K_{\ddot{q}}q^2 \\ &\quad + Y_{\ddot{r}}\ddot{v} + K_{\ddot{r}}\ddot{p} + N_{\ddot{r}}\ddot{r} - X_{\ddot{v}}v^2 - X_{\ddot{r}}vr - (X_{\ddot{p}} - Y_{\ddot{q}})vp + M_{\ddot{r}}rp + K_{\ddot{p}}p^2 \\ &\quad - (X_{\ddot{u}} - Y_{\ddot{v}})uv - X_{\ddot{w}}vw + (X_{\ddot{q}} + Y_{\ddot{p}})up + Y_{\ddot{r}}ur + Z_{\ddot{q}}wp \\ &\quad - (X_{\ddot{q}} + Y_{\ddot{p}})vq - (K_{\ddot{p}} - M_{\ddot{q}})pq - K_{\ddot{r}}qr \end{aligned}$$

Coriolis–Centripetal Forces in a Rotating Coordinate System

The added mass Coriolis and centripetal matrix is found by collecting all terms in Kirchhoff's equations that are not functions of body accelerations.

Property 6.2 (Added-mass Coriolis-centripetal matrix) For a rigid-body moving through an ideal fluid the hydrodynamic Coriolis and centripetal matrix can always be parameterized such that it is skew-symmetric

$$\mathbf{C}_A(\boldsymbol{\nu}) = -\mathbf{C}_A^\top(\boldsymbol{\nu}), \quad \forall \boldsymbol{\nu} \in \mathbb{R}^6$$

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) \\ -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{A}_{21}\boldsymbol{\nu}_1 + \mathbf{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix}$$

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_3 & a_2 \\ 0 & 0 & 0 & a_3 & 0 & -a_1 \\ 0 & 0 & 0 & -a_2 & a_1 & 0 \\ 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \end{bmatrix}$$

$$a_1 = X_{\dot{u}}u + X_{\dot{v}}v + X_{\dot{w}}w + X_{\dot{p}}p + X_{\dot{q}}q + X_{\dot{r}}r$$

$$a_2 = Y_{\dot{u}}u + Y_{\dot{v}}v + Y_{\dot{w}}w + Y_{\dot{p}}p + Y_{\dot{q}}q + Y_{\dot{r}}r$$

$$a_3 = Z_{\dot{u}}u + Z_{\dot{v}}v + Z_{\dot{w}}w + Z_{\dot{p}}p + Z_{\dot{q}}q + Z_{\dot{r}}r$$

$$b_1 = K_{\dot{u}}u + K_{\dot{v}}v + K_{\dot{w}}w + K_{\dot{p}}p + K_{\dot{q}}q + K_{\dot{r}}r$$

$$b_2 = M_{\dot{u}}u + M_{\dot{v}}v + M_{\dot{w}}w + M_{\dot{p}}p + M_{\dot{q}}q + M_{\dot{r}}r$$

$$b_3 = N_{\dot{u}}u + N_{\dot{v}}v + N_{\dot{w}}w + N_{\dot{p}}p + N_{\dot{q}}q + N_{\dot{r}}r$$

Sagatun, S. I. and T. I. Fossen (1991). Lagrangian Formulation of Underwater Vehicles' Dynamics. *IEEE International Conference on Systems, Man, and Cybernetics*, Charlottesville, VA, USA, IEEE Xplore, pp. 1029-1034.

MSS Toolbox

```
function C = m2c(M,nu)
% C = m2c(M,nu) computes the Coriolis-centripetal matrix C(nu) from the
% the system inertia matrix M > 0 for varying velocity nu.
% If M is a 6x6 matrix and nu = [u, v, w, p, q, r]', the output is a 6x6 C matrix
% If M is a 3x3 matrix and nu = [u, v, r]', the output is a 3x3 C matrix.
%
% Examples: CRB = m2c(MRB,nu)
%           CA = m2c(MA, nu)
% Output:
%   C: Coriolis-centripetal matrix C = C(nu)
%
% Inputs:
%   M: 6x6 or 3x3 rigid-body MRB or added mass MA system matrix
%   nu: nu = [u, v, w, p, q, r]' or nu = [u, v, r]'
%
% The Coriolis and centripetal matrix depends on nu1 = [u,v,w]' and nu2 =
% [p,q,r]' as shown in Fossen (2021, Theorem 3.2). Alternatively, the matrix
% CRB = CRB(nu2) can be computed using
%
% [MRB,CRB] = rbody(m,R44,R55,R66,nu2,r_bp)

M = 0.5 * (M + M'); % Symmetrization of the inertia matrix

if (length(nu) == 6) % 6-DOF model
M11 = M(1:3,1:3); M12 = M(1:3,4:6); M21 = M12'; M22 = M(4:6,4:6);

nu1 = nu(1:3);
nu2 = nu(4:6);
nu1_dot = M11 * nu1 + M12 * nu2;
nu2_dot = M21 * nu1 + M22 * nu2;

C = [ zeros(3,3)  -Smtrx(nu1_dot)
      -Smtrx(nu1_dot) -Smtrx(nu2_dot) ];

else % 3-DOF model (surge, sway and yaw)
C = [ 0 0 -M(2,2)*nu(2)-M(2,3)*nu(3)
      0 0 M(1,1)*nu(1)
      M(2,2)*nu(2)+M(2,3)*nu(3) -M(1,1)*nu(1) 0 ];

end
```

Note that $\mathbf{C}_{RB}$ and $\mathbf{C}_A$ can be computed automatically from $\mathbf{M}_{RB}$ and $\mathbf{M}_A$ both for 3-DOF and 6-DOF models.

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) \\ -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{A}_{21}\boldsymbol{\nu}_1 + \mathbf{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix}$$

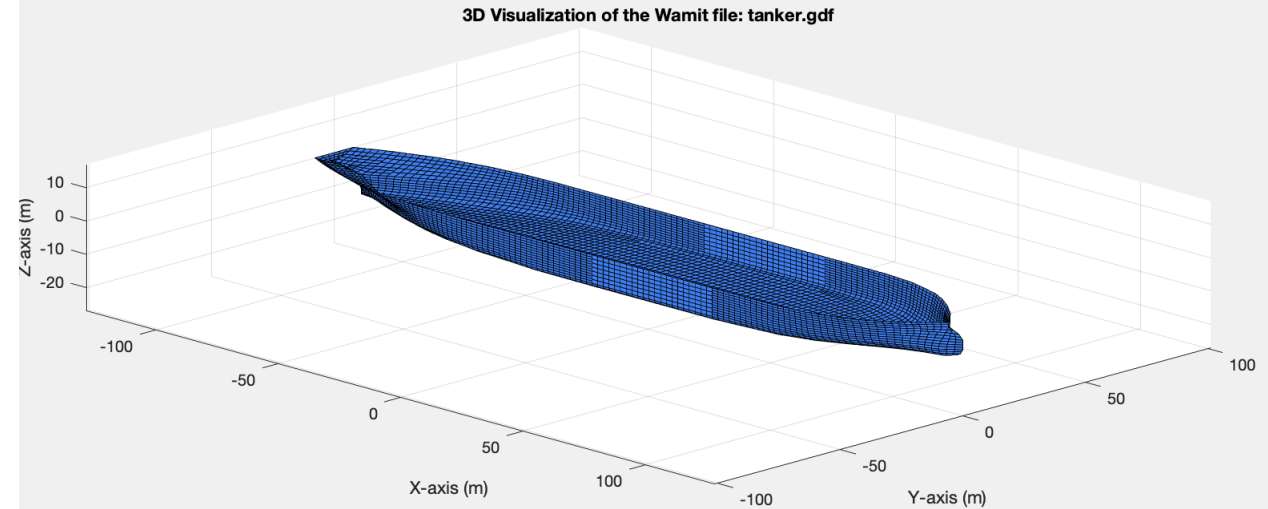
$$\mathbf{M}_A = \mathbf{M}_A^\top = - \begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \quad (N_{\dot{v}} = Y_{\dot{r}})$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = -\mathbf{C}_A^\top(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

MSS Toolbox

```
>> load tanker
>> vessel
vessel =
  struct with fields:
    main: [1x1 struct]
    velocities: 0
    headings: [1x36 double]
    MRB: [6x6 double]
    C: [6x6x60 double]
    freqs: [1x60 double]
    A: [6x6x60 double]
    B: [6x6x60 double]
    motionRAO: [1x1 struct]
    forceRAO: [1x1 struct]
    driftfr: [1x1 struct]
    Bv: [6x6 double]
>> vessel.freqs
ans =
Columns 1 through 8
  0    0.0383    0.0424    0.0468    0.0517    0.0571    0.0631    0.0697
Columns 9 through 16
  0.0770    0.0850    0.0939    0.1037    0.1145    0.1264    0.1395    0.1540
Columns 17 through 24
  0.1700    0.1876    0.2070    0.2284    0.2519    0.2779    0.3063    0.3378
Columns 25 through 32
  0.3722    0.4101    0.4517    0.4971    0.5473    0.6018    0.6621    0.7281
Columns 33 through 40
  0.7994    0.8775    0.9622    1.0560    1.1571    1.2668    1.3840    1.5104
Columns 41 through 48
  1.6491    1.8003    1.9574    2.1299    2.3100    2.5033    2.7083    2.9361
Columns 49 through 56
  3.1574    3.3963    3.6530    3.9270    4.1888    4.4880    4.7600    5.0671
Columns 57 through 60
  5.3702    5.7120    5.9840    10.0000
```

```
>> plot_wamitgdf('tanker')
```



```
>> MA = vessel.A(:,:,1) Zero-frequency added mass matrix
MA =
  1.0e+11 *
    0.0000    0    -0.0000    0    -0.0128    0
    0    0.0005    0    0.0030    0    0.0029
   -0.0000    0    0.0039    0    0.0096    0
    0    0.0030    0    0.1103    0    -0.0419
   -0.0128    0    0.0096    0    9.0021    0
    0    0.0029    0    -0.0420    0    1.7383
```

```
>> nu = [10 0 0 0 0 0]';
>> CA = m2c(MA,nu)
CA =
  1.0e+10 *
    0    0    0    0    -0.0032    0
    0    0    0    0.0032    0    0.0041
    0    0    0    0    -0.0041    0
    0    -0.0032    0    0    0    1.2758
    0.0032    0    0.0041    0    0    0
    0    -0.0041    0    -1.2758    0    0
```

Coriolis–Centripetal Forces in a Rotating Coordinate System

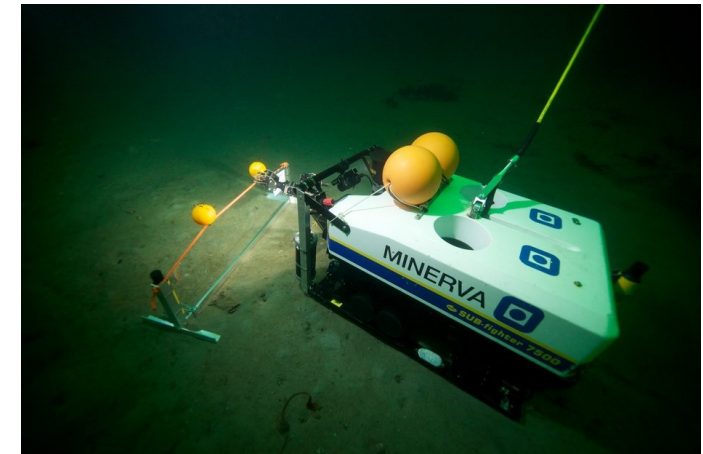
Example 6.2 (Added mass for low-speed AUVs and ROVs)

In general, the motion of an underwater vehicle moving in 6 DOFs at high speed is highly nonlinear and coupled. However, in many AUV and ROV applications, the vehicle operates at low speed. If the vehicle also has three planes of symmetry and the body-fixed frame is located at the CG, the off-diagonal elements of the added-mass matrix $\mathbf{M}_A$ are usually small and may therefore be neglected. Hence, the following simplified expressions for $\mathbf{M}_A$ and $\mathbf{C}_A(\boldsymbol{\nu}_r)$ are obtained:

$$\mathbf{M}_A = \mathbf{M}_A^\top = -\text{diag}\{X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}\}$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = -\mathbf{C}_A^\top(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}}w_r & Y_{\dot{v}}v_r \\ 0 & 0 & 0 & Z_{\dot{w}}w_r & 0 & -X_{\dot{u}}u_r \\ 0 & 0 & 0 & -Y_{\dot{v}}v_r & X_{\dot{u}}u_r & 0 \\ 0 & -Z_{\dot{w}}w_r & Y_{\dot{v}}v_r & 0 & -N_{\dot{r}}r & M_{\dot{q}}q \\ Z_{\dot{w}}w_r & 0 & -X_{\dot{u}}u_r & N_{\dot{r}}r & 0 & -K_{\dot{p}}p \\ -Y_{\dot{v}}v_r & X_{\dot{u}}u_r & 0 & -M_{\dot{q}}q & K_{\dot{p}}p & 0 \end{bmatrix}.$$

The diagonal approximation is widely used because it is time consuming to determine the off-diagonal elements experimentally or theoretically. In practice, the approximation is often quite accurate since the off-diagonal elements of a positive inertia matrix are typically much smaller than the corresponding diagonal elements.



Coriolis–Centripetal Forces in a Rotating Coordinate System

Example 6.1 (Added mass for ships and USVs)

For surface ships and USVs, it is common to decouple the surge mode from the steering dynamics due to symmetry about the xz plane. Similarly, the heave, pitch, and roll modes are neglected under the assumption that these motions are small. Hence, for $\boldsymbol{\nu}_r = [u_r, v_r, r]^\top$ the added-mass matrix and added-mass Coriolis–centripetal matrix become

$$\mathbf{M}_A = \mathbf{M}_A^\top = - \begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \quad (N_{\dot{v}} = Y_{\dot{r}})$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = -\mathbf{C}_A^\top(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}.$$

The Coriolis and centripetal forces corresponding to (6.42) are

$$\mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \begin{bmatrix} Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r \\ \underbrace{(X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r}_{\text{Munk moment}} \end{bmatrix}$$

where the first term in the yaw moment is the nonlinear Munk moment, which is known to have destabilizing effects.



Nonlinear Damping Forces

Potential theory codes assume that the fluid is inviscid. Water is assumed to be an **inviscid fluid**, in which the **viscosity of the fluid is equal to zero**. When viscous forces are neglected, such as the case of inviscid flow, the Navier-Stokes equation can be simplified to a form known as the Euler equation.

Consequently, potential codes only compute:

- ✓ **Potential Damping:** Dissipative force in an inviscid fluid. The contribution from the potential damping terms compared to other dissipative terms like viscous damping is usually negligible.

The Hydrodynamic Paradox

D'Alembert's paradox (or the hydrodynamic paradox), first formulated in 1752 by Jean le Rond d'Alembert, states that in incompressible, inviscid, and **irrotational potential flow**, the drag force on a body moving at constant velocity relative to the fluid is zero.

This theoretical result contradicts experimental observations, where bodies moving in fluids such as air or water experience significant drag, particularly at high Reynolds numbers.

Therefore, **viscous damping forces, such as quadratic drag and lift, must be included in seakeeping and maneuvering models in addition to potential damping.**



Jean le Rond d'Alembert
(1717-1783)

Nonlinear Damping Forces

Sources of Viscous Damping

✓ Skin friction

- Boundary layer shear stresses on the hull.
- Small effect on oscillatory motions but always present.

✓ Eddy and vortex shedding

- Flow separation at bilge keels, chines, appendages.
- Dominant contribution to roll damping.

✓ Lift-induced damping

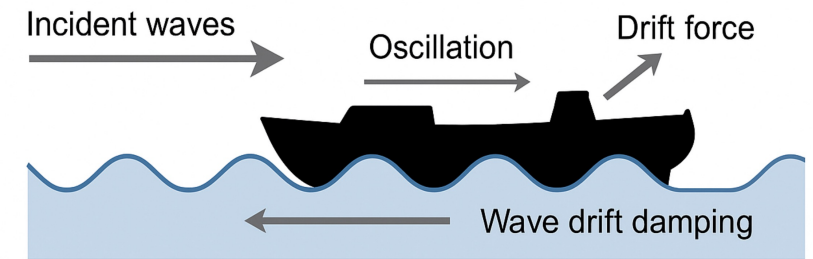
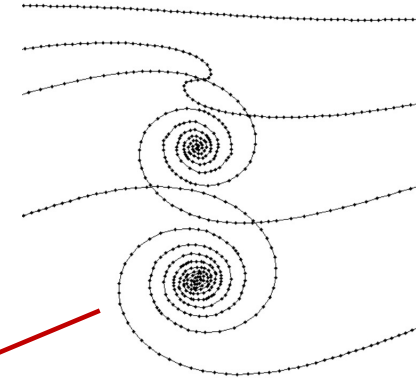
- Fins, rudders, and appendages generating side forces.
- Important for control and stabilization.

✓ Wave drift damping

- Interaction of mean drift forces with oscillations.
- Often treated in second-order wave load analysis.

✓ Other nonlinear viscous effects

- Flow separation, turbulence, appendage–hull interaction.
- Typically lumped into empirical coefficients.



- When a vessel oscillates in waves, it experiences order steady drift force (mean wave drift).
- If the vessel is moving relative to this drift, part of the drift force acts as a resisting force : wave drift damping.
- This effect grows with wave height H_s & depends on heading, frequency, and hull form.

Unfortunately, it is very hard to model all these effects and collect the terms into a common nonlinear damping matrix $\mathbf{D}(\mathbf{v})$. Linear damping must be blended with the nonlinear damping terms to avoid “double counting) at higher speed , e.g., using $\mathbf{D} \exp(-\alpha U_g) \mathbf{v} + \mathbf{D}(\mathbf{v}) \mathbf{v}$.

Nonlinear Damping Forces

Property 6.3 (Hydrodynamic damping)

The damping matrix $\mathbf{D}(\boldsymbol{\nu}_r)$ and the nonlinear damping vector $\mathbf{d}(\boldsymbol{\nu}_r)$ satisfy

$$\mathbf{x}^\top \mathbf{D}(\boldsymbol{\nu}_r) \mathbf{x} > 0, \quad \forall \mathbf{x} \neq \mathbf{0}$$

and

$$\boldsymbol{\nu}_r^\top \mathbf{d}(\boldsymbol{\nu}_r) \geq 0, \quad \forall \boldsymbol{\nu}_r \neq \mathbf{0}.$$

Hence, the total damping force satisfies

$$\boldsymbol{\nu}_r^\top (\mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)) > 0, \quad \forall \boldsymbol{\nu}_r \neq \mathbf{0}$$

so that mechanical energy is always dissipated.

$$\underbrace{\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{\mathbf{M}_A \dot{\boldsymbol{\nu}}_r + \mathbf{C}_A(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

The vector $\mathbf{d}(\boldsymbol{\nu}_r)$ is nonlinear dissipative forces and moments not expressible in the form $\mathbf{D}(\boldsymbol{\nu}_r) \boldsymbol{\nu}_r$

Property 6.3 applies for linear and nonlinear damping.

Linear Damping

For slender bodies like ships and USVs, it is common to decouple the surge velocity from the other DOFs. Hence, we can express the damping matrix $\mathbf{D}$ as a function of the hydrodynamic derivatives:

$$\mathbf{D} = - \begin{bmatrix} X_u & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_v & 0 & Y_p & 0 & Y_r \\ 0 & 0 & Z_w & 0 & Z_q & 0 \\ 0 & K_v & 0 & K_p & 0 & K_r \\ 0 & 0 & M_w & 0 & M_q & 0 \\ 0 & N_v & 0 & N_p & 0 & N_r \end{bmatrix}$$

We can estimate a constant 6x6 damping matrix $\mathbf{D} = \mathbf{B}^{\text{eq}} + \mathbf{B}_v$ from power-averaged seakeeping coefficients or time constants/relative damping ratios.

Nonlinear Surge Damping

ITTC Resistance

$$X = -\frac{1}{2}\rho S(1+k)C_f(u_r)|u_r|u_r$$

$$C_f(u_r) = \underbrace{\frac{0.075}{(\log_{10}(R_n) - 2)^2}}_{C_F} + C_R$$

where

- ρ density of water
- S wetted surface of the hull
- k form factor giving a viscous correction (typically 0.1 for ships in transit)
- C_F flat plate friction from the ITTC (1957) line
- C_R residual friction due to hull roughness, pressure resistance, wave making resistance

Reynolds number:

$$R_n = \frac{L_{pp}}{\nu}|u_r| \geq 0 \quad \nu = 1 \times 10^{-6} \text{ m}^2/\text{s} \text{ is the kinematic viscosity at } 20^\circ \text{ C.}$$

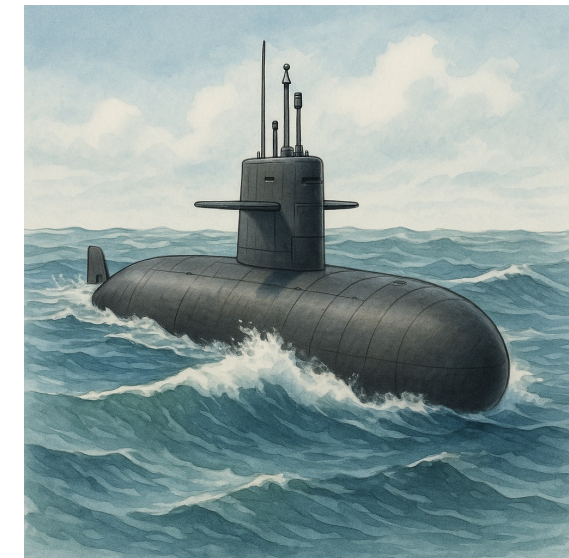
Notation of SNAME (1950)

$$X = X_{|u|u}|u_r|u_r$$

$$X_{|u|u} = -\frac{1}{2}\rho S(1+k)C_f(u_r) < 0$$

$$u_r = u - u_c$$

$$= u - V_c \cos(\beta_{V_c} - \psi)$$



For DP applications, the ITTC formulas often give too little damping compared to what is observed in model tests. Hence, an additional linear damping term is usually included at low speed. At higher speeds, the quadratic drag term becomes dominant and the linear damping contribution should gradually fade out. Hence,

$$X = e^{-k_u|u_r|}X_u u_r + X_{|u|u}|u_r|u_r \quad k_u = 3-5,$$

Cross-Flow Drag Principle

For relative current angles $|\beta_{vc} - \psi| \gg 0$, where β_{vc} is the ocean current direction, the cross-flow drag principle may be applied to calculate the nonlinear damping force in sway and the yaw moment (Faltinsen, 1990)

$$Y = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L_{pp}}{2}}^{\frac{L_{pp}}{2}} T(x) |v_r + xr| (v_r + xr) dx,$$

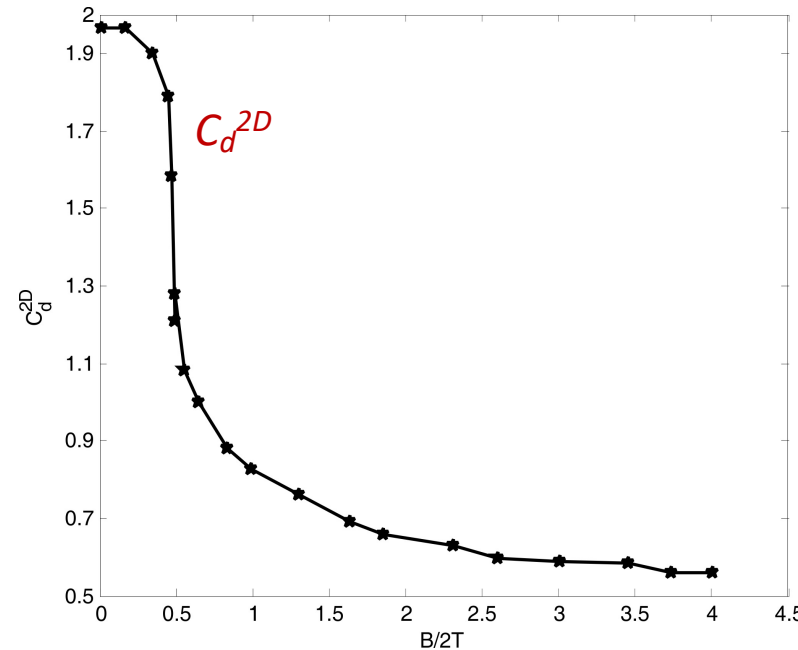
$$N = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L_{pp}}{2}}^{\frac{L_{pp}}{2}} T(x) x |v_r + xr| (v_r + xr) dx$$

where C_d^{2D} is the 2-D drag coefficient and $T(x)$ is the draft.

Strip theory means that the ship is cut in section along the x-axis. The integral is evaluated by summing up the contribution from each section.

Extension to 6-DOF Cross-Flow Drag

$$d(\nu_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr| (v_r + xr) \\ T(x) |w_r + xq| (w_r + xq) \\ 0 \\ x T(x) |w_r + xq| (w_r + xq) \\ x T(x) |v_r + xr| (v_r + xr) \end{bmatrix} dx$$



Dr. Sighard F. Hoerner (1906-1971) is the author of two famous compendiums in aerodynamics entitled, Fluid-Dynamic Drag and Fluid-Dynamic Lift.

Matlab/GNU Octave:

The 2-D drag coefficient C_d^{2D} can be estimated as a function of beam B and draft T using Hoerner's curve. This is implemented in the MSS toolbox by `Cd = Hoerner(B,T)`. The function

```
tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model)
```

computes the 6-DOF cross-flow drag integral using 20 strip sections. The default setting is `drag_model = 'Hoerner'`. Alternatively, `'cylinder'` uses the DNV recommended practice for environmental conditions and environmental loads to compute C_d^{2D} for a cylinder-shaped vehicle.

MSS Toolbox

```
>> L = 20; B = 8; T = 5;
>> nu_r = [1 0.1 0.2 0.1 0.1 -0.1]';
>> tau_crossflow = crossFlowDrag(L,B,T,nu_r,'Hoerner')
tau_crossflow =
    1.0e+05 *
         0
    -0.0510
    -0.1030
         0
    -1.4979
     1.4222
>> tau_crossflow = crossFlowDrag(L,B,T,nu_r,'cylinder')
tau_crossflow =
    1.0e+04 *
         0
    -0.3022
    -0.6099
         0
    -8.8728
     8.4242
```

The “**cylinder**” option is intended for cylindrical AUVs, using cylinder data from DNV-RP-C205. Conventional ship hulls are instead more accurately represented by Hoerner’s curve.

```
function tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model)
% tau_crossflow = crossFlowDrag(L,B,T,nu_r,drag_model) computes the cross-flow
% drag integrals for a marine craft using strip theory. Application:
%
%   M * d/dt nu_r + C(nu_r)*nu_r + D*nu_r + g(eta) = tau + tau_crossflow
%
% Inputs: L:   length
%          B:   beam
%          T:   draft
%          nu_r = [u-u_c, v-v_c, w-w_c, p, q, r]': Relative velocity vector
%          drag_model (Optional) : 'Hoerner' (default), 'cylinder'
%
% Output: tau_crossflow = [0 Yh Zh 0 Mh Nh]: 6-DOF cross-flow drag vector
```

```
rho = 1025; % Density of water
dx = L / 20; % Divide hull into 20 strips

switch drag_model
case 'Hoerner'
    % 2-D drag coefficient based on Hoerner's curve
    Cd_2D = Hoerner(B,T);
case 'cylinder'
    % 2D drag coefficient based on cylinder data from DNV-RP-C205
    Cd_2D = cylinderDrag(L,B,nu_r);
otherwise
    error('Unsupported drag model %s.',drag_model)
end

Yh = 0; Zh = 0; Mh = 0; Nh = 0;
for xL = -L/2:dx:L/2
    v_r = nu_r(2); % Relative sway velocity
    w_r = nu_r(3); % Relative heave velocity
    q = nu_r(5); % Pitch rate
    r = nu_r(6); % Yaw rate
    U_h = abs(v_r + xL * r) * (v_r + xL * r);
    U_v = abs(w_r + xL * q) * (w_r + xL * q);
    Yh = Yh - 0.5 * rho * T * Cd_2D * U_h * dx; % Sway force
    Zh = Zh - 0.5 * rho * T * Cd_2D * U_v * dx; % Heave force
    Mh = Mh - 0.5 * rho * T * Cd_2D * xL * U_v * dx; % Pitch moment
    Nh = Nh - 0.5 * rho * T * Cd_2D * xL * U_h * dx; % Yaw moment
end

tau_crossflow = [0 Yh Zh 0 Mh Nh]';
```

Nonlinear Roll, Pitch and Yaw Damping

The surge resistance and cross-flow drag model may be supplemented by additional nonlinear damping terms in roll, pitch, and yaw. This is often necessary to capture viscous and vortex-induced damping effects that are not well represented by linear hydrodynamic damping alone, in particular for roll motion of ships and offshore vessels.

In yaw, the nonlinear damping term is often required to counteract the destabilizing Munk moment arising from the added-mass Coriolis and centripetal matrix $\mathbf{C}_A(\mathbf{v}_r)$. At moderate and high forward speed, the **Munk moment** may generate an unstable yawing moment, especially for slender ships and underwater vehicles.

$$\begin{aligned}
 K &= K_p (1 + \kappa_4 |p|) p \\
 M &= M_q (1 + \kappa_5 |q|) q \\
 N &= N_r (1 + \kappa_6 |r|) r
 \end{aligned}$$

Coefficient	Mode	Units	Typical Range
κ_4	Roll	s	1–10
κ_5	Pitch	s	0.5–5
κ_6	Yaw	s	0.5–3



Linear and Nonlinear Hydrodynamic Damping

Hydrodynamic damping (Sections 6.1.1 and 6.1.3):

$$\mathbf{D}(\boldsymbol{\nu}_r) = - \begin{bmatrix} X_u^* & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p^* & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q^* & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r^* \end{bmatrix}, \quad \begin{aligned} X_u^* &= e^{-k_u |u_r|} X_u + X_{|u|u} |u_r| \\ K_p^* &= K_p (1 + \kappa_4) |p| \\ M_q^* &= M_q (1 + \kappa_5) |q| \\ N_r^* &= N_r (1 + \kappa_6) |r| \end{aligned}$$

$$\mathbf{d}(\boldsymbol{\nu}_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr|(v_r + xr) \\ T(x) |w_r + xq|(w_r + xq) \\ 0 \\ x T(x) |w_r + xq|(w_r + xq) \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx$$

Strip theory cross-flow drag integrals using a 2-D drag coefficient.

The exponential factor $e^{-k_u |u_r|}$ fades the linear surge damping coefficient X_u with increasing speed to avoid double counting of surge drag when quadratic damping $X_{|u|u} |u_r|$ is included.

The nominal linear roll, pitch, and yaw damping derivatives are augmented by amplitude-dependent damping through the coefficients κ_4 , κ_5 , and κ_6 .

Alternatively, when maneuvering derivatives are unavailable, the linear damping coefficients $X_u, Y_v, Z_w, K_p, M_q, N_r$, etc. may be estimated from frequency-domain sea-keeping coefficients by power-based averaging (Section 5.4.2). This yields an equivalent constant linear damping matrix $\mathbf{B}_0^{\text{eq}}$, which can be combined with viscous damping contributions $\mathbf{B}_v$ to obtain $\mathbf{D} \approx \mathbf{B}_0^{\text{eq}} + \mathbf{B}_v$.



Summary: First-Principles 6-DOF Maneuvering Model

Two state-space formulations are commonly used. The first uses the **relative velocity** $\mathbf{v}_r$ as state and is convenient when ocean currents are modeled explicitly and relative velocity is measured (e.g., by a DVL). The second uses the **absolute velocity** $\mathbf{v}$ as state and is often preferred for surface vessels, where INS/GNSS systems provide measurements of absolute velocity.

Independent of $\mathbf{v}_1 = [u, v, w]^T$

$$\underbrace{M_{RB}\dot{\mathbf{v}} + C_{RB}(\mathbf{v})\mathbf{v}}_{\text{Rigid-body forces}} + \underbrace{M_A\dot{\mathbf{v}}_r + C_A(\mathbf{v}_r)\mathbf{v}_r + D(\mathbf{v}_r)\mathbf{v}_r + d(\mathbf{v}_r)}_{\text{Hydrodynamic forces}} + \underbrace{g(\boldsymbol{\eta})}_{\text{Hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\begin{aligned} M &= M_{RB} + M_A \\ C(\mathbf{v}_r) &= C_{RB}(\mathbf{v}_r) + C_A(\mathbf{v}_r) \end{aligned}$$

Relative velocity vector

$$\mathbf{v}_r = \mathbf{v} - \mathbf{v}_c$$

Irrotational ocean current

$$\mathbf{v}_c = \underbrace{[u_c, v_c, w_c, 0, 0, 0]^T}_{\mathbf{v}_c^b}$$

$$\mathbf{v}_c^b = [u_c, v_c, w_c]^T$$

$$\dot{\mathbf{v}}_c = \begin{bmatrix} -S(\boldsymbol{\omega}_{nb}^b)\mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Relative-velocity state-space model:

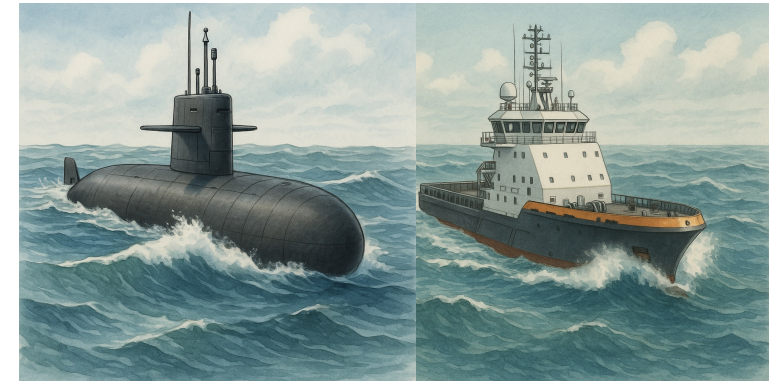
$$\dot{\boldsymbol{\eta}} = J_{\Theta}(\boldsymbol{\eta})(\mathbf{v}_r + \mathbf{v}_c)$$

$$\dot{\mathbf{v}}_r = M^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - C(\mathbf{v}_r)\mathbf{v}_r - D(\mathbf{v}_r)\mathbf{v}_r - d(\mathbf{v}_r) - g(\boldsymbol{\eta}))$$

Absolute-velocity state-space model:

$$\dot{\boldsymbol{\eta}} = J_{\Theta}(\boldsymbol{\eta})\mathbf{v}$$

$$\dot{\mathbf{v}} = \dot{\mathbf{v}}_c + M^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - C(\mathbf{v}_r)\mathbf{v}_r - D(\mathbf{v}_r)\mathbf{v}_r - d(\mathbf{v}_r) - g(\boldsymbol{\eta}))$$



$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{Rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + C_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{Hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{Hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

Rigid-body matrices (Section 3.3.2)

$$M_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bG}^b) \\ m\mathbf{S}(\mathbf{r}_{bG}^b) & \mathbf{I}_b^b \end{bmatrix}$$

$$C_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bG}^b) \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_b^b \boldsymbol{\nu}_2) \end{bmatrix} \mathbf{H}(\mathbf{r}_{bG}^b) \quad (\text{independent of } \boldsymbol{\nu}_1).$$

Added-mass matrices (Section 6.1.1)

$$M_A = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix}$$

$$C_A(\boldsymbol{\nu}_r) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{12}\boldsymbol{\nu}_{r,2}) \\ -\mathbf{S}(\mathbf{A}_{11}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{12}\boldsymbol{\nu}_{r,2}) & -\mathbf{S}(\mathbf{A}_{21}\boldsymbol{\nu}_{r,1} + \mathbf{A}_{22}\boldsymbol{\nu}_{r,2}) \end{bmatrix}$$

- Generalized position vector: $\boldsymbol{\eta} = [x^n, y^n, z^n, \phi, \theta, \psi]^\top$.
- Generalized velocity vector: $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$.
- Ocean current velocity vector: $\boldsymbol{\nu}_c = [u_c, v_c, w_c, 0, 0, 0]^\top$.
- Relative velocity vector: $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$.

$$\begin{aligned} \boldsymbol{\nu}_{r,1} &= [u_r, v_r, w_r]^\top \\ \boldsymbol{\nu}_{r,2} &= \boldsymbol{\nu}_2 = [p, q, r]^\top \end{aligned} \quad \dot{\boldsymbol{\nu}}_c = \begin{bmatrix} -\mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix}$$

Hydrodynamic damping (Sections 6.1.1 and 6.1.3)

$$D(\boldsymbol{\nu}_r) = - \begin{bmatrix} X_u^* & X_v & X_w & X_p & X_q & X_r \\ Y_u & Y_v & Y_w & Y_p & Y_q & Y_r \\ Z_u & Z_v & Z_w & Z_p & Z_q & Z_r \\ K_u & K_v & K_w & K_p^* & K_q & K_r \\ M_u & M_v & M_w & M_p & M_q^* & M_r \\ N_u & N_v & N_w & N_p & N_q & N_r^* \end{bmatrix}, \quad \begin{aligned} X_u^* &= e^{-k_u|u_r|} X_u + X_{|u|u}|u_r| \\ K_p^* &= K_p (1 + \kappa_4) |p| \\ M_q^* &= M_q (1 + \kappa_5) |q| \\ N_r^* &= N_r (1 + \kappa_6) |r| \end{aligned}$$

$$\mathbf{d}(\boldsymbol{\nu}_r) = -\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |v_r + xr|(v_r + xr) \\ T(x) |w_r + xq|(w_r + xq) \\ 0 \\ x T(x) |w_r + xq|(w_r + xq) \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx$$

Hydrostatic forces (Sections 4.1.1 and 4.2.1):

Neutrally buoyant submarines, ROVs and AUVs:

$$\mathbf{g}(\boldsymbol{\eta}) = [0, 0, 0, (z_G - z_B)W \cos(\theta) \sin(\phi), (z_G - z_B)W \sin(\theta), 0]^T$$

Ships and USVs:

$$\mathbf{g}(\boldsymbol{\eta}) \approx \mathbf{G}\boldsymbol{\eta}$$

where

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho g A_{wp} & 0 & -\rho g A_{wp} x_F & 0 \\ 0 & 0 & 0 & WGM_T & 0 & 0 \\ 0 & 0 & -\rho g A_{wp} x_F & 0 & \rho g A_{wp} x_F^2 + WGM_L & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\underbrace{\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{\mathbf{M}_A\dot{\boldsymbol{\nu}}_r + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{\mathbf{g}(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$



Special Cases and Simplified Maneuvering Models

Longitudinal and Lateral Maneuvering Model for Marine Craft with Weak Cross-Coupling

Ships and USVs. For ships and USVs operating in waves, the added-mass and damping matrices are commonly approximated by constant equivalent matrices obtained from the frequency-dependent seakeeping coefficients:

$$\mathbf{M}_A = \mathbf{A}_0^{\text{eq}}, \quad \mathbf{D} = \mathbf{B}_0^{\text{eq}} + \mathbf{B}_v.$$

Underwater vehicles. For underwater vehicles operating below the wave-affected zone, the surrounding fluid may be regarded as unbounded. In this case, the hydrodynamic coefficients are independent of the wave-excitation frequency, so that

$$\begin{aligned} \mathbf{M}_A &= \mathbf{A}(\omega) = \text{constant} \quad \forall \omega, \\ \mathbf{D} &= \mathbf{B}_v, \quad \mathbf{B}(\omega) = \mathbf{0}. \end{aligned}$$



Longitudinal and Lateral Maneuvering Model for Marine Craft with Weak Cross-Coupling

Assumption 6.1 (Weak Longitudinal-Lateral Coupling)

The equations of motion are expressed in a body-fixed frame $\{b\}$ with origin at the CG. The craft is symmetric with respect to the xz-plane, and hydrodynamic coupling between longitudinal and lateral motions is neglected. The state vectors are

$$\begin{aligned}\boldsymbol{\eta}^{(\text{lon})} &= [x^n, z^n, \theta]^\top, & \boldsymbol{\nu}^{(\text{lon})} &= [u, w, q]^\top \\ \boldsymbol{\eta}^{(\text{lat})} &= [y^n, \phi, \psi]^\top, & \boldsymbol{\nu}^{(\text{lat})} &= [v, p, r]^\top.\end{aligned}$$

This approximation is often appropriate for slender marine vehicles operating near straight-line motion, including submarines, torpedo-shaped AUVs, and conventional monohull ships, where longitudinal and lateral dynamics are only weakly coupled.

The resulting 3-DOF **longitudinal** and **lateral models** on the next slides are based on the following assumptions and simplifications:

Body-fixed coordinate origin: The equations are expressed in a body-fixed frame with **origin at the CG**. This eliminates the rigid-body coupling terms associated with $\mathbf{x}_G$, $\mathbf{y}_G$, and $\mathbf{z}_G$.

Diagonal approximation: In high-fidelity maneuvering models, the off-diagonal added-mass and damping terms are usually retained. However, for slender vehicles with weak cross-coupling, and especially when only limited hydrodynamic data are available, it is common to approximate the **added-mass** and **linear damping matrices** by their diagonal elements. Expressing the equations at the CG reduces the neglected off-diagonal terms.

Longitudinal Equations of Motion based on Relative Velocity

$$\begin{aligned}
 \begin{bmatrix} \dot{x}^n \\ \dot{z}^n \\ \dot{\theta} \end{bmatrix} &= \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \left(\begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} + \begin{bmatrix} u_c \\ w_c \\ 0 \end{bmatrix} \right) \\
 \begin{bmatrix} m - X_{\dot{u}} & 0 & 0 \\ 0 & m - Z_{\dot{w}} & 0 \\ 0 & 0 & I_y - M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \dot{u}_r \\ \dot{w}_r \\ \dot{q} \end{bmatrix} \\
 + \begin{bmatrix} 0 & 0 & (m - Z_{\dot{w}})w_r \\ 0 & 0 & -(m - X_{\dot{u}})u_r \\ -(m - Z_{\dot{w}})w_r & (m - X_{\dot{u}})u_r & 0 \end{bmatrix} \begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} \\
 + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & \rho g A_{wp} & -\rho g A_{wp} x_F \\ 0 & -\rho g A_{wp} x_F & \rho g A_{wp} x_F^2 + WGM_L \end{bmatrix}}_{\text{For submerged vehicles, replace by } [0, -z_B W \sin(\theta), 0]^T} \begin{bmatrix} x^n \\ z^n \\ \theta \end{bmatrix} \\
 + \begin{bmatrix} -e^{-k_u |u_r|} X_u - X_{|u|u} |u_r| & 0 & 0 \\ 0 & -Z_w & 0 \\ 0 & 0 & -M_q (1 + \kappa_5 |q|) \end{bmatrix} \begin{bmatrix} u_r \\ w_r \\ q \end{bmatrix} \\
 - \underbrace{\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} 0 \\ T(x) |w_r + xq| (w_r + xq) \\ x T(x) |w_r + xq| (w_r + xq) \end{bmatrix} dx}_{\text{Optional crossflow drag}} = \begin{bmatrix} \tau_1 \\ \tau_3 \\ \tau_5 \end{bmatrix} + \begin{bmatrix} \tau_{\text{wind},1} \\ \tau_{\text{wind},3} \\ \tau_{\text{wind},5} \end{bmatrix} + \begin{bmatrix} \tau_{\text{wave},1} \\ \tau_{\text{wave},3} \\ \tau_{\text{wave},5} \end{bmatrix}
 \end{aligned}$$



Lateral Equations of Motion based on Relative Velocity

$$\begin{aligned}
 \begin{bmatrix} \dot{y}^n \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} &= \begin{bmatrix} \cos(\psi) & 0 & 0 \\ 0 & 1 & \sin(\phi) \tan(\theta) \\ 0 & 0 & \cos(\phi) / \cos(\theta) \end{bmatrix} \left(\begin{bmatrix} v_r \\ p \\ r \end{bmatrix} + \begin{bmatrix} v_c \\ 0 \\ 0 \end{bmatrix} \right) \\
 \begin{bmatrix} m - Y_{\dot{v}} & 0 & 0 \\ 0 & I_x - K_{\dot{p}} & 0 \\ 0 & 0 & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{v}_r \\ \dot{p} \\ \dot{r} \end{bmatrix} \\
 + \begin{bmatrix} 0 & -(m - Z_{\dot{w}})w_r & (m - X_{\dot{u}})u_r \\ (m - Z_{\dot{w}})w_r & 0 & 0 \\ -(m - X_{\dot{u}})u_r & 0 & 0 \end{bmatrix} \begin{bmatrix} v_r \\ p \\ r \end{bmatrix} \\
 + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & WGM_T & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y^n \\ \phi \\ \psi \end{bmatrix}}_{\text{For submerged vehicles, replace by } [0, -z_B W \sin(\phi), 0]^T} \\
 + \begin{bmatrix} -Y_v & 0 & 0 \\ 0 & -K_p(1 + \kappa_4|p|) & 0 \\ 0 & 0 & -N_r(1 + \kappa_6|r|) \end{bmatrix} \begin{bmatrix} v_r \\ p \\ r \end{bmatrix} \\
 - \underbrace{\frac{\rho C_d^{2D}}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \begin{bmatrix} T(x) |v_r + xr|(v_r + xr) \\ 0 \\ x T(x) |v_r + xr|(v_r + xr) \end{bmatrix} dx}_{\text{Optional crossflow drag}} = \begin{bmatrix} \tau_2 \\ \tau_4 \\ \tau_6 \end{bmatrix} + \begin{bmatrix} \tau_{\text{wind},2} \\ \tau_{\text{wind},4} \\ \tau_{\text{wind},6} \end{bmatrix} + \begin{bmatrix} \tau_{\text{wave},2} \\ \tau_{\text{wave},4} \\ \tau_{\text{wave},6} \end{bmatrix}
 \end{aligned}$$



Horizontal-Plane Maneuvering Model for Ships and USVs with Port–Starboard Symmetry

Assumption 6.2 (Horizontal-plane motion)

Assume port–starboard symmetry such that $y_G = 0$, and choose the body-fixed frame $\{b\}$ so that the CG is located at $(x_G, 0, 0)$. Furthermore, assume that heave, roll, and pitch motions are small and may be neglected. Under these assumptions, the craft motion is approximated by the horizontal-plane states (x^n, y^n, ψ) and (u, v, r) .



Nonlinear Equations of Motion based on Relative Velocity

$$\underbrace{\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{\mathbf{M}_A\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r}_{\text{hydrodynamic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi)\boldsymbol{\nu}$$

The nonlinear hydrodynamic derivatives in sway and yaw can also be approximated by cross-flow drag integrals

$$\begin{bmatrix} Y_{|v|v}|v_r| + Y_{|r|v}|r| & Y_{|v|r}|v_r| + Y_{|r|r}|r| \\ N_{|v|v}|v_r| + N_{|r|v}|r| & N_{|v|r}|v_r| + N_{|r|r}|r| \end{bmatrix} \begin{bmatrix} v_r \\ r \end{bmatrix} \approx -\frac{\rho C_d^{2D}}{2} \int_{-L/2}^{L/2} \begin{bmatrix} T(x)|v_r + xr|(v_r + xr) \\ xT(x)|v_r + xr|(v_r + xr) \end{bmatrix} dx$$



$$\mathbf{M}_{RB} = \begin{bmatrix} m & 0 & 0 \\ 0 & m & mx_G \\ 0 & mx_G & I_z \end{bmatrix}, \quad \mathbf{M}_A = -\begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & N_{\dot{v}} & N_{\dot{r}} \end{bmatrix}$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & -m(x_G r + v) \\ 0 & 0 & mu \\ m(x_G r + v) & -mu & 0 \end{bmatrix}$$

$$\mathbf{C}_A(\boldsymbol{\nu}_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

$$\mathbf{D}(\boldsymbol{\nu}_r) = -\begin{bmatrix} e^{-k_u|u_r|}X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix}$$

$$-\begin{bmatrix} X_{|u|u}|u_r| & 0 & 0 \\ 0 & Y_{|v|v}|v_r| + Y_{|r|v}|r| & Y_{|v|r}|v_r| + Y_{|r|r}|r| \\ 0 & N_{|v|v}|v_r| + N_{|r|v}|r| & N_{|v|r}|v_r| + N_{|r|r}|r| \end{bmatrix}$$

Horizontal-Plane Maneuvering Model for Ships and USVs with Port–Starboard Symmetry

Since $\mathbf{C}_{RB}(\boldsymbol{\nu})$ is parametrized independent of linear velocity it follows from Property 10.1 that

$$\mathbf{M}\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$

$$\mathbf{M} = \mathbf{M}_A + \mathbf{M}_{RB}$$

$$\mathbf{C}(\boldsymbol{\nu}_r) = \mathbf{C}_A(\boldsymbol{\nu}_r) + \mathbf{C}_{RB}(\boldsymbol{\nu}_r).$$

State-Space Models for Horizontal-Plane Motions

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi) \left(\boldsymbol{\nu}_r + \begin{bmatrix} u_c \\ v_c \\ 0 \end{bmatrix} \right)$$

$$\dot{\boldsymbol{\nu}}_r = \mathbf{M}^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r)$$

$$u_c = V_c \cos(\beta_{V_c} - \psi)$$

$$v_c = V_c \sin(\beta_{V_c} - \psi)$$

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi)\boldsymbol{\nu}$$

$$\dot{\boldsymbol{\nu}} = \begin{bmatrix} rv_c \\ -ru_c \\ 0 \end{bmatrix} + \mathbf{M}^{-1}(\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} - \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r - \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r)$$

where $\boldsymbol{\nu}_r = \boldsymbol{\nu} - [u_c, v_c, 0]^\top$.

$$\dot{u}_c = rV_c \sin(\beta_{V_c} - \psi) = rv_c$$

$$\dot{v}_c = -rV_c \cos(\beta_{V_c} - \psi) = -ru_c$$

Longitudinal and Lateral Maneuvering Model for Marine Craft with Weak Cross-Coupling

Several linear and nonlinear maneuvering models are included the MSS toolbox.

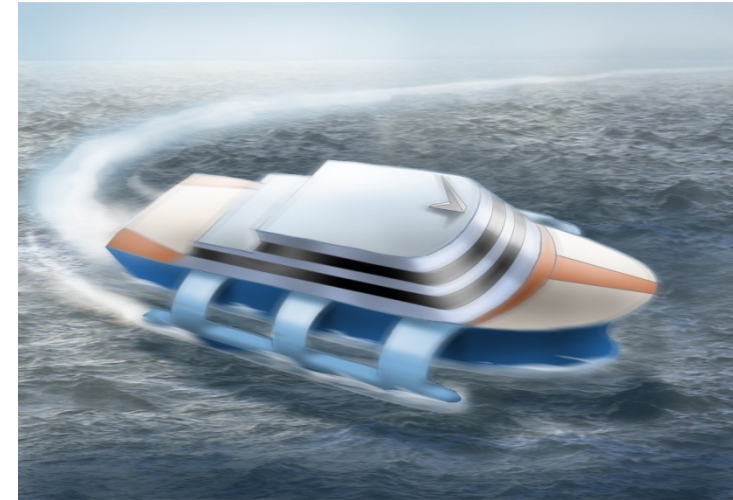
Matlab/GNU Octave:

Several MSS maneuvering models are located in the toolbox catalog
`../MSS/CRAFT/.`

```
[xdot,U] = container(x,tau)    % Container ship L = 175 m
[xdot,U] = Lcontainer(x,tau)   % Linearized container ship
[xdot,U] = mariner(x,ui,U0)    % Mariner class vessel L = 161 m
[xdot,U] = navalvessel(x,tau)  % Naval vessel L = 51.5 m
[xdot,U] = tanker(x,ui)        % Tanker L = 304.8 m
[xdot,U,M,B_p] = otter(x,n,mp,rp,V_c,beta_c) % Otter USV L = 2 m
```

The models can be simulated under feedback control using the user editable scripts

`Sim<model name>.m`



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MSS Toolbox

Computation of Ocean Current Velocities in BODY and NED

We assume that the current is irrotational and constant in NED. Consequently,

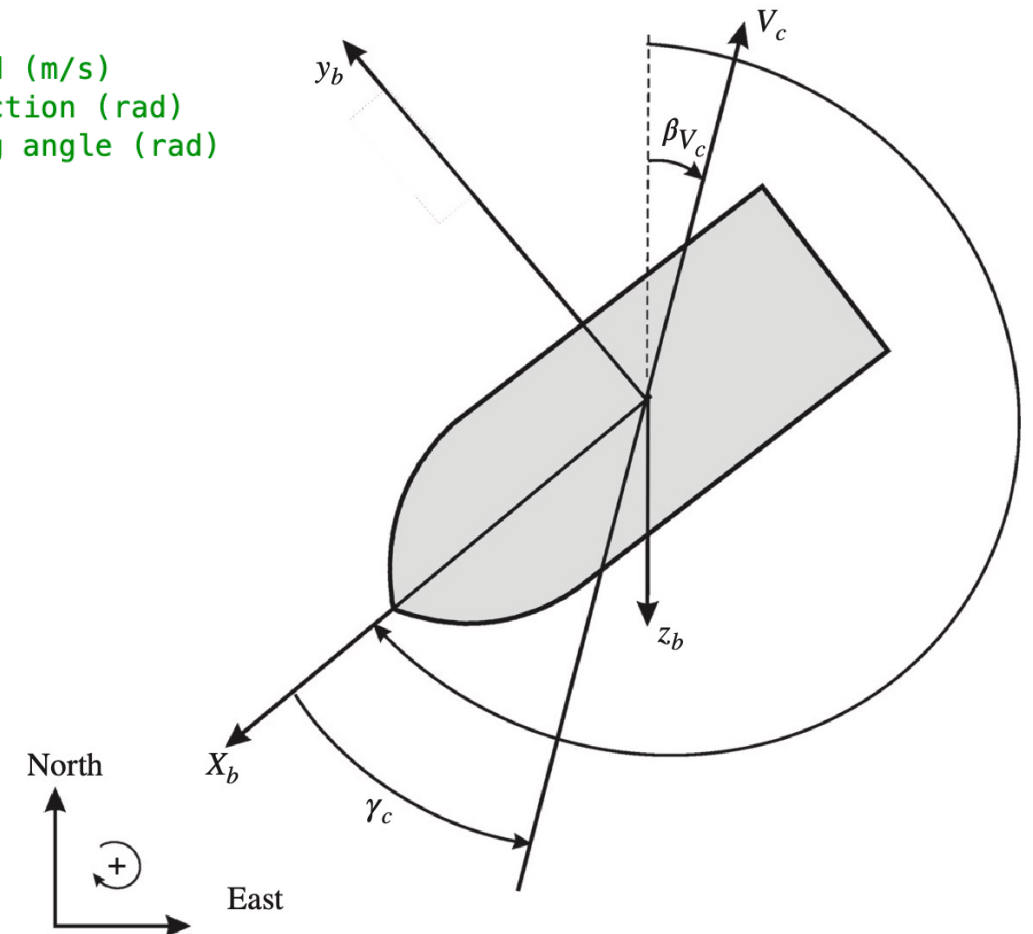
```
% constant 2-D ocean current expressed in NED
Vc = 1; % current speed (m/s)
betaVc = deg2rad(30); % current direction (rad)
psi = deg2rad(10); % craft heading angle (rad)
```

```
% current velocities expressed in NED
vc_n = [Vc * cos(betaVc) Vc * sin(betaVc) 0]'
```

```
% current velocities expressed in BODY
vc_b = Rzyx(0,0,psi)' * vc_n
```

```
vc_n =
    0.8660
    0.5000
         0
vc_b =
    0.9397
    0.3420
         0
```

```
>> help Rzyx
R = Rzyx(phi,theta,psi) computes the Euler angle
rotation matrix R in SO(3) using the zyx convention
```



3-DOF Linear Maneuvering Model for Ships and USVs

Linear Maneuvering Equations (Surge–Sway–Yaw)

$$\underbrace{(M_{RB} + M_A)}_M \dot{\nu}_r + \underbrace{(C_{RB} + C_A + D)}_N \nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$



$$C_{RB} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & mU_0 \\ 0 & 0 & mx_G U_0 \end{bmatrix}, \quad C_A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -X_{\dot{u}} U_0 \\ 0 & (X_{\dot{u}} - Y_{\dot{v}}) U_0 & -Y_{\dot{r}} U_0 \end{bmatrix}$$

$$D = - \begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix}$$

U_0 is the cruise speed used for linearization

$$\begin{bmatrix} m - X_{\dot{u}} & 0 & 0 \\ 0 & m - Y_{\dot{v}} & mx_G - Y_{\dot{r}} \\ 0 & mx_G - N_{\dot{v}} & I_z - N_{\dot{r}} \end{bmatrix} \begin{bmatrix} \dot{u}_r \\ \dot{v}_r \\ \dot{r} \end{bmatrix} + \begin{bmatrix} -X_u & 0 & 0 \\ 0 & -Y_v & (m - X_{\dot{u}})U_0 - Y_r \\ 0 & (X_{\dot{u}} - Y_{\dot{v}})U_0 - N_v & (mx_G - Y_{\dot{r}})U_0 - N_r \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ r \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_6 \end{bmatrix}$$

For a rudder operating in the propeller slipstream,

$$\tau_1 = -X_{\delta\delta}\delta^2 + (1 - t)T_p, \quad \tau_2 = -Y_{\delta}\delta, \quad \tau_6 = -N_{\delta}\delta$$



3-DOF Linear Maneuvering Model for Ships and USVs

Decoupling the surge-sway-yaw model into a surge subsystem and a sway-yaw subsystem:

Surge Subsystem

$$(m - X_{\dot{u}})\dot{u}_r - X_u u_r = X_{\delta\delta}\delta^2 + (1 - t_p)T_p$$

Sway-Yaw Subsystem

$$X_u = \frac{8A_1}{3\pi} X_{|u|u}$$

$$\mathbf{M}\dot{\boldsymbol{\nu}}_r + \mathbf{N}\boldsymbol{\nu}_r = \mathbf{b}\delta$$

$$\mathbf{M} = \begin{bmatrix} m - Y_{\dot{v}} & mx_G - Y_{\dot{r}} \\ mx_G - N_{\dot{v}} & I_z - N_{\dot{r}} \end{bmatrix}$$

$$\mathbf{N} = \begin{bmatrix} -Y_v & (m - X_{\dot{u}})U_0 - Y_r \\ (X_{\dot{u}} - Y_{\dot{v}})U_0 - N_v & (mx_G - Y_{\dot{r}})U_0 - N_r \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} -Y_{\delta} \\ -N_{\delta} \end{bmatrix}.$$



Notice: $\mathbf{C}_A(\mathbf{v})$ includes the famous destabilizing **Munk moment** (from aerodynamics) and some other $\mathbf{C}_A$ -terms

3-DOF Linear Models for Dynamic Positioning (DP)

DP models are derived under the assumption of low-speed operation (typically up to 2m/s) , such that the quadratic Coriolis and centripetal terms and the nonlinear damping forces are negligible compared to the linear damping terms.

Kinematic Nonlinearity

$R(\psi(t)) := R(t)$ (The nonlinearity associated with the rotation matrix may be removed by treating the **yaw angle** as **measured**)

Linear Time-Varying (LTV) Model for DP Controller-Observer Design

$$\begin{aligned}\dot{\eta} &= R(t)\nu \\ M\dot{\nu} + D\nu &= R^\top(t)b + \tau + \tau_{\text{wind}} + \tau_{\text{wave}} \\ \dot{b} &= 0 \quad (\text{The bias } b \text{ is constant in NED})\end{aligned}$$

$$\tau = Bu \quad (\text{Control forces and moments})$$

Matlab / GNU Octave:

The MSS DP models are located in the catalog `../MSS/CRAFT/SHIPS/`.

```
[xdot,U,M] = osv(x,ui,Vc,betaVc) % Supply vessel L = 83.0 m
[xdot,U,M,D] = supply(x, tau)    % Supply vessel L = 76.2 m
```

The models can be simulated under feedback control using the user editable scripts

`Sim<model name>.m`

Linear Time-Varying (LTV) State-Space Model

$$\dot{x} = \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 3} & R(t) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -M^{-1}D & M^{-1}R^\top(t) \\ \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} \end{bmatrix}}_{A(t)} x + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times r} \\ M^{-1}B \\ \mathbf{0}_{3 \times r} \end{bmatrix}}_B u + \underbrace{\begin{bmatrix} \mathbf{0}_{3 \times 3} \\ M^{-1} \\ \mathbf{0}_{3 \times 3} \end{bmatrix}}_E w$$

$$x = [\eta^\top, \nu^\top, b^\top]^\top$$

$$w = \tau_{\text{wind}} + \tau_{\text{wave}}$$



Classical Maneuvering Models

The Abkowitz (1964) Maneuvering Model

Data-Driven Maneuvering Modeling

Previous maneuvering models were derived from first principles:

- Surge resistance
- Cross-flow drag
- Hydrodynamic force models

An alternative approach is to fit experimental maneuvering data directly.

Abkowitz's Idea (1964)

Represent the surge force, sway force, and yaw moment by a truncated

Taylor-series expansion about straight-line motion:

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\boldsymbol{\tau}_{RB} = [X(\mathbf{x}), Y(\mathbf{x}), N(\mathbf{x})]^\top$$

$$X(\mathbf{x}) \approx X(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial X(\mathbf{x})}{\partial x_i} \bigg|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 X(\mathbf{x})}{(\partial x_i)^2} \bigg|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 X(\mathbf{x})}{(\partial x_i)^3} \bigg|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

$$Y(\mathbf{x}) \approx Y(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial Y(\mathbf{x})}{\partial x_i} \bigg|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 Y(\mathbf{x})}{(\partial x_i)^2} \bigg|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 Y(\mathbf{x})}{(\partial x_i)^3} \bigg|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

$$N(\mathbf{x}) \approx Z(\mathbf{x}_0) + \sum_{i=1}^n \left(\frac{\partial N(\mathbf{x})}{\partial x_i} \bigg|_{\mathbf{x}_0} \Delta x_i + \frac{1}{2} \frac{\partial^2 N(\mathbf{x})}{(\partial x_i)^2} \bigg|_{\mathbf{x}_0} \Delta x_i^2 + \frac{1}{6} \frac{\partial^3 N(\mathbf{x})}{(\partial x_i)^3} \bigg|_{\mathbf{x}_0} \Delta x_i^3 \right)$$

Key Assumptions

1. Third-order Taylor expansion about ($\mathbf{u} = \mathbf{U}_0$).
2. Acceleration terms appear only linearly.
3. Port–starboard symmetry is enforced.
4. Acceleration–velocity coupling terms are neglected.

$$\mathbf{x} = [u, v, r, \dot{u}, \dot{v}, \dot{r}, \delta]^\top$$

$$\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0 = [\Delta x_1, \Delta x_2, \dots, \Delta x_7]^\top$$

$$\mathbf{x}_0 = [U, 0, 0, 0, 0, 0, 0]^\top$$

The Abkowitz (1964) Maneuvering Model

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\boldsymbol{\tau}_{RB} = [X(\mathbf{x}), Y(\mathbf{x}), N(\mathbf{x})]^\top$$

Taylor-Series Expansion

$$\begin{aligned} X = & X^* + X_{\dot{u}}\dot{u} + X_u\Delta u + X_{uu}\Delta u^2 + X_{uuu}\Delta u^3 + X_{vv}v^2 + X_{rr}r^2 + X_{\delta\delta}\delta^2 \\ & + X_{rv\delta}rv\delta + X_{r\delta}r\delta + X_{v\delta}v\delta + X_{vvu}v^2\Delta u + X_{rru}r^2\Delta u + X_{\delta\delta u}\delta^2\Delta u \\ & + X_{rvu}rvu + X_{r\delta u}r\delta\Delta u + X_{v\delta u}v\delta\Delta u \end{aligned}$$

$$\begin{aligned} Y = & Y^* + Y_u\Delta u + Y_{uu}\Delta u^2 + Y_rr + Y_vv + Y_{\dot{r}}\dot{r} + Y_{\dot{v}}\dot{v} + Y_{\delta}\delta + Y_{rrr}r^3 + Y_{vvv}v^3 \\ & + Y_{\delta\delta\delta}\delta^3 + Y_{rr\delta}r^2\delta + Y_{\delta\delta r}\delta^2r + Y_{rrv}r^2v + Y_{vvv}v^2r + Y_{\delta\delta v}\delta^2v + Y_{vv\delta}v^2\delta \\ & + Y_{\delta vr}\delta vr + Y_{vu}v\Delta u + Y_{vuu}v\Delta u^2 + Y_{ru}r\Delta u + Y_{ruu}r\Delta u^2 + Y_{\delta u}\delta\Delta u \\ & + Y_{\delta uu}\delta\Delta u^2 \end{aligned}$$

$$\begin{aligned} N = & N^* + N_u\Delta u + N_{uu}\Delta u^2 + N_rr + N_vv + N_{\dot{r}}\dot{r} + N_{\dot{v}}\dot{v} + N_{\delta}\delta + N_{rrr}r^3 \\ & + N_{vvv}v^3 + N_{\delta\delta\delta}\delta^3 + N_{rr\delta}r^2\delta + N_{\delta\delta r}\delta^2r + N_{rrv}r^2v + N_{vvv}v^2r + N_{\delta\delta v}\delta^2v \\ & + N_{vv\delta}v^2\delta + N_{\delta vr}\delta vr + N_{vu}v\Delta u + N_{vuu}v\Delta u^2 + N_{ru}r\Delta u + N_{ruu}r\Delta u^2 \\ & + N_{\delta u}\delta\Delta u + N_{\delta uu}\delta\Delta u^2 \end{aligned}$$



Second-Order Modulus Model (Fedyaevsky-Sobolev Model)

The idea of using **second-order modulus functions** to represent the nonlinear dissipative forces and moment originates from the work of Fedyaevsky and Sobolev (1963). The model is particularly suitable for maneuvering at moderate and high speeds, where the hydrodynamic loads are dominated by quadratic cross-flow drag and propulsion losses. This modeling philosophy was adopted by Norrbin (1970), who formulated a nonlinear maneuvering model based on modulus-function representations of the hydrodynamic forces and moments.

$$M\dot{\nu}_r + C(\nu_r)\nu_r + D(\nu_r)\nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

$$\begin{aligned} & C_A(\nu_r)\nu_r + D(\nu_r)\nu_r \\ &= \begin{bmatrix} Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r \\ (X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r \end{bmatrix} \left(\text{alternatively} \begin{bmatrix} X_{vr}v_r r + X_{rr}r^2 \\ Y_{ur}u_r r \\ N_{uv}u_r v_r + N_{ur}u_r r \end{bmatrix} \right) \\ &+ \begin{bmatrix} -X_{|u|u}|u_r|u_r \\ -Y_{|v|v}|v_r|v_r - Y_{|v|r}|v_r|r - Y_{v|r}|v_r|r| - Y_{|r|r}|r|r| \\ -N_{|v|v}|v_r|v_r - N_{|v|r}|v_r|r - N_{v|r}|v_r|r| - N_{|r|r}|r|r| \end{bmatrix} \\ &= \begin{bmatrix} -X_{|u|u}|u_r|u_r + Y_{\dot{v}}v_r r + Y_{\dot{r}}r^2 \\ -X_{\dot{u}}u_r r - Y_{|v|v}|v_r|v_r - Y_{|v|r}|v_r|r - Y_{v|r}|v_r|r| - Y_{|r|r}|r|r| \\ (X_{\dot{u}} - Y_{\dot{v}})u_r v_r - Y_{\dot{r}}u_r r - N_{|v|v}|v_r|v_r \\ -N_{|v|r}|v_r|r - N_{v|r}|v_r|r| - N_{|r|r}|r|r| \end{bmatrix} \end{aligned}$$



¹The $C_A(\nu_r)\nu_r$ terms may alternatively be written as $X_{vr}v_r r$, $X_{rr}r^2$, $Y_{ur}u_r r$, $N_{uv}u_r v_r$, and $N_{ur}u_r r$. If these coefficients are obtained experimentally, they generally contain viscous contributions in addition to the potential-flow coefficients $Y_{\dot{v}}$, $X_{\dot{u}}$, and $Y_{\dot{r}}$.

Second-Order Modulus Model (Blanke, 1981)

$$M\dot{\nu}_r + C(\nu_r)\nu_r + D(\nu_r)\nu_r = \tau + \tau_{\text{wind}} + \tau_{\text{wave}}$$

$$C_A(\nu_r) = \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u_r \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r & 0 \end{bmatrix}$$

$$D(\nu_r) = \begin{bmatrix} -X_{|u|u}|u_r| & 0 & 0 \\ 0 & -Y_{|v|v}|v_r| - Y_{|r|v}|r| & -Y_{|v|r}|v_r| - Y_{|r|r}|r| \\ 0 & -N_{|v|v}|v_r| - N_{|r|v}|r| & -N_{|v|r}|v_r| - N_{|r|r}|r| \end{bmatrix}$$

For large ships, Blanke (1981) suggested a simplified damping model which neglects the $|r|$ -dependent terms. Hence,

$$C_A(\nu_r) + D(\nu_r) = \begin{bmatrix} -X_{|u|u}|u_r| & 0 & Y_{\dot{v}}v_r + Y_{\dot{r}}r \\ 0 & -Y_{|v|v}|v_r| & -X_{\dot{u}}u_r - Y_{|v|r}|v_r| \\ -Y_{\dot{v}}v_r - Y_{\dot{r}}r & X_{\dot{u}}u_r - N_{|v|v}|v_r| & -N_{|v|r}|v_r| \end{bmatrix}$$

A simplified version of Norrbin's model, retaining only the most important terms for steering and propulsion-loss prediction, was proposed by Blanke (1981). This model may be interpreted as a 2nd-order approximation of the cross-flow drag integrals.



MMG Standard Maneuvering Model

The Maneuvering Modeling Group (MMG) standard model, developed by the Japanese MMG committee and later standardized by Yasukawa and Yoshimura (2015) is an empirical 3-DOF maneuvering model for ships in surge, sway, and yaw. Unlike the second-order modulus and Abkowitz models, where the hydrodynamic forces and yaw moment are represented directly by polynomial or modulus functions of the body-fixed velocities, the MMG model decomposes the total force and moment into separate contributions from the hull, propeller, and rudder. This decomposition gives the model a clear physical interpretation and makes it possible to combine captive model tests with empirical formulas for the propeller and rudder.

$$\mathbf{M}\dot{\boldsymbol{\nu}}_r + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \boldsymbol{\tau}_H + \boldsymbol{\tau}_P + \boldsymbol{\tau}_R + \boldsymbol{\tau}_{\text{env}}$$

$$\begin{bmatrix} X \\ Y \\ N \end{bmatrix} = \begin{bmatrix} X_H + X_P + X_R \\ Y_H + Y_R \\ N_H + N_R \end{bmatrix}$$

The hull force and moment contributions are represented by nonlinear hydrodynamic derivatives.

A common MMG representation is

$$\begin{aligned} X_H &= X_{uu}u_r^2 + X_{vv}v_r^2 + X_{vr}v_r r + X_{rr}r^2 \\ Y_H &= Y_v v_r + Y_r r + Y_{|v|v}|v_r|v_r + Y_{|r|r}|r|r + Y_{vr}v_r r \\ N_H &= N_v v_r + N_r r + N_{|v|v}|v_r|v_r + N_{|r|r}|r|r + N_{vr}v_r r \end{aligned}$$

Propulsion force (Section 9.1)

$$X_P = (1 - t_p) T_p, \quad T_p = \rho D_p^4 K_T(J_a) |n_p| n_p$$

Rudder force (Section 9.5)

$$\begin{aligned} X_R &= -(1 - t_R) F_N \sin(\delta) \\ Y_R &= -(1 + a_H) F_N \cos(\delta) \\ N_R &= -(x_R + a_H x_H) F_N \cos(\delta) \end{aligned}$$

$$F_N = \frac{1}{2} \rho A_R f_\alpha U_R^2 \sin(\alpha_R)$$

Chapter Goals - Revisited

- Formulate the **first-principles 6-DOF maneuvering model** by combining rigid-body kinetics with hydrodynamic added mass, damping, hydrostatic restoring, and external forces.
- Explain how the rigid-body and added-mass **Coriolis–centripetal matrices** arise in **BODY-fixed coordinates** and state their skew-symmetry property.
- Explain when **zero-frequency potential coefficients** are appropriate for surge, sway, and yaw, why **viscous damping must be added**, and why the approximation fails for coupled 6-DOF motion.
- Apply the **power-based spectral averaging method** to compute sea-state-dependent constant equivalent added-mass and damping matrices for **6-DOF maneuvering models**.
- Model linear and nonlinear hydrodynamic damping using **ITTC surge resistance, cross-flow drag**, and additional **roll, pitch, and yaw damping**.
- Include **constant irrotational ocean currents** using relative velocity and formulate the dynamics with either relative or absolute velocity as the state.
- Derive and compare **simplified and classical maneuvering models**, including longitudinal–lateral, horizontal-plane 3-DOF, dynamic positioning, Abkowitz, second-order modulus, and MMG models.

$$\underbrace{M_{RB}\dot{\boldsymbol{\nu}} + C_{RB}(\boldsymbol{\nu})\boldsymbol{\nu}}_{\text{rigid-body forces}} + \underbrace{M_A\dot{\boldsymbol{\nu}}_r + C_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + D(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{d}(\boldsymbol{\nu}_r)}_{\text{hydrodynamic forces}} + \underbrace{g(\boldsymbol{\eta})}_{\text{hydrostatic forces}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}}$$